AI & COMPUTING

Elements of Plane Trigonometry

Hugh Blackburn

FAHAD.COM — Free Book Library

Public Domain Edition

Project Gutenberg’s Elements of Plane Trigonometry, by Hugh Blackburn This eBook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org Title: Elements of Plane Trigonometry For the use of the junior class of mathematics in the University of Glasgow Author: Hugh Blackburn Release Date: June 25, 2010 [EBook #32973] Language: English Character set encoding: ISO-8859-1 *** START OF THIS PROJECT GUTENBERG EBOOK ELEMENTS OF PLANE TRIGONOMETRY ***

Produced by Andrew D. Hwang, Laura Wisewell and the Online Distributed Proofreading Team at http://www.pgdp.net (The original copy of this book was generously made available for scanning by the Department of Mathematics at the University of Glasgow.) transcriber’s note Minor typographical corrections and presentational changes have been made without comment. Figures may have been moved slightly with respect to the surrounding text. This PDF file is formatted for screen viewing, but may be easily formatted for printing. Please consult the preamble of the LATEX source file for instructions.

ELEMENTS OF PLANE TRIGONOMETRY FOR THE USE OF THE JUNIOR CLASS OF MATHEMATICS IN THE UNIVERSITY OF GLASGOW. BY HUGH BLACKBURN, M.A. PROFESSOR OF MATHEMATICS IN THE UNIVERSITY OF GLASGOW, LATE FELLOW OF TRINITY COLLEGE, CAMBRIDGE. Lon˘n and New York: MACMILLAN AND CO. □□□□. [All Rights reserved .]

Cambridge: PRINTED BY C. J. CLAY, M.A. AT THE UNIVERSITY PRESS.

PREF ACE. Some apology is required for adding another to the long list of books on Trigonometry. My excuse is that during twenty years’ experi- ence I have not found any published book exactly suiting the wants of my Students. In conducting a Junior Class by regular progressive steps from Euclid and Elementary Algebra to Trigonometry, I have had to fill up by oral instruction the gap between the Sixth Book of Euclid and the circular measurement of Angles; which is not satisfactorily bridged by the propositions of Euclid’s Tenth and Twelfth Books usually sup- posed to be learned; nor yet by demonstrations in the modern books on Trigonometry, which mostly follow Woodhouse; while the Appen- dices to Professor Robert Simson’s Euclid in the editions of Professors Playfair and Wallace of Edinburgh, and of Professor James Thomson of Glasgow, seemed to me defective for modern requirements, as not sufficiently connected with Analytical Trigonometry. What I felt the want of was a short Treatise, to be used as a Text Book after the Sixth Book of Euclid had been learned and some knowl- edge of Algebra acquired, which should contain satisfactory demon- strations of the propositions to be used in teaching Junior Students the Solution of Triangles, and should at the same time lay a solid founda- tion for the study of Analytical Trigonometry. This want I have attempted to supply by applying, in the first Chap- ter, Newton’s Method of Limits to the mensuration of circular arcs and areas; choosing that method both because it is the strictest and the easiest, and because I think the Mathematical Student should be early introduced to the method. The succeeding Chapters are devoted to an exposition of the nature of the Trigonometrical ratios, and to the demonstration by geometrical constructions of the

principal propositions required for the Solution of Triangles. To these I have added a general explanation of the appli- cations of these propositions in Trigonometrical Surveying: and I have iii

TRIGONOMETRY. iv concluded with a proof of the formulæ for the sine and cosine of the sum of two angles treated (as it seems to me they should be) as ex- amples of the Elementary Theory of Projection. Having learned thus much the Student has gained a knowledge of Trigonometry as origi- nally understood, and may apply his knowledge in Surveying; and he has also reached a point from which he may advance into Analytical Trigonometry and its use in Natural Philosophy. Thinking that others may have felt the same want as myself, I have published the Tract instead of merely printing it for the use of my Class. H. B.

ELEMENTS OF PLANE TRIGONOMETRY. Trigonometry (from trÐgwnon , triangle, and metrèw , I measure) is the science of the numerical relations between the sides and angles of triangles. This Treatise is intended to demonstrate, to those who have learned the principal propositions in the first six books of Euclid, so much of Trigonometry as was originally implied in the term, that is, how from given values of some of the sides and angles of a triangle to calculate, in the most convenient way, all the others. A few propositions supplementary to Euclid are premised as intro- ductory to the propositions of Trigonometry as usually understood. CHAPTER I. OF THE MENSURATION OF THE CIRCLE. Def. . A magnitude or ratio, which is fixed in value by the con- ditions of the question, is called a Constant. Def. . A magnitude or ratio, which is not fixed in value by the conditions of the question and which is conceived to change its value by lapse of time, or otherwise, is called a Variable. Def. . If a variable shall be always less than a given constant, but shall in time become greater than any less constant, the given constant is the Superior Limit of the variable: and if the variable shall be

[Chap. I.] TRIGONOMETRY. always greater than a given constant but in time shall become less than any greater constant, the given constant is the Inferior Limit of the variable. Lemma. If two variables are at every instant equal their limits are equal. For if the limits be not equal, the one variable shall necessarily in time become greater than the one limit and less than the other, while at the same instant the other variable shall be greater than both limits or less than both limits, which is impossible, since the variables are always equal. Def. . Curvilinear segments are similar when, if on the chord of the one as base any triangle be described with its vertex in the arc, a similar triangle with its vertex in the other arc can always be described on its chord as base; and the arcs are Similar Curves. Cor. . Arcs of circles subtending equal angles at the centres are similar curves. Cor. . If a polygon of any number of sides be inscribed in one of two similar curves, a similar polygon can be inscribed in the other. Def. . Let a number of points be taken in a terminated curve line, and let straight lines be drawn from each point to the next, then if the number of points be conceived to increase and the distance between each two to diminish continually, the extremities remaining fixed, the limit of the sum of the straight lines is called the Length of the Curve. Prop. I. The lengths of similar arcs are proportional to their chords. For let any number of points be taken in the one and the points be joined by

straight lines so as to inscribe a polygon in it, and let a similar polygon be inscribed in the other, the perimeters of the two polygons are proportional to the chords, or the ratio of the perimeter of the one

[Chap. I.] OF THE MENSURATION OF THE CIRCLE. □ to its chord is equal to the ratio of the perimeter of the other to its chord. Then if the number of sides of the polygons increase these two ratios vary but remain always equal to each other, therefore (Lemma) their limits are equal. But the limit of the ratio of the perimeter of the polygon to the chord is (Def. □) the ratio of the length of the curve to its chord, therefore the ratio of the length of the one curve to its chord is equal to the ratio of the length of the other curve to its chord, or the lengths of similar finite curve lines are proportional to their chords. Cor. □. Since semicircles are similar curves and the diameters are their chords, the ratio of the semi-circumference to the diameter is the same for all circles. If this ratio be denoted, as is customary, by π 2 , then numerically the circumference÷ the diameter = π, and the circumference = 2πR. O A B C Cor. □. The angle subtended at the centre of a circle by an arc equal to the radius is the same for all circles. For if AC be the arc equal to the radius, and AB the arc subtending a right an- gle, then by Euclid vi. □□ AOC :AOB ::AC :AB. But AB is a fourth of the circumfer- ence = πR 2 ; therefore AOC : a right angle :: R : πR 2 :: 2 :π or numerically AOC = 2 π× a right angle,

[Chap. I.] TRIGONOMETRY. □ that is the angle subtended by an arc equal to the radius is a fixed fraction of a right angle. Prop. II. The areas of similar segments are proportional to the squares on their chords. For, if similar polygons of any number of sides be inscribed in the similar segments, they are to one another in the duplicate ratio of the chords, or, alternately, the ratio of the polygon inscribed in the one segment to the square on its chord is the same as the ratio of the similar polygon in the other segment to the square on its chord. Now conceive the polygons to vary by the number of sides increasing continually while the two polygons remain always similar, then the variable ratios of the polygons to the squares on the chords always remain equal, and therefore their limits are equal (Lemma); and these limits are obviously the ratios of the areas of the segments to the squares on the chords, which ratios are therefore equal. Cor. Circles are to one another as the squares of their diameters. Note. From Prop. II. and III. it is obvious that “The correspond- ing sides, whether straight or curved, of similar figures, are proportion- als; and their areas are in the duplicate ratio of the sides.” (Newton, Princip. I. Sect. i. Lemma v.) Prop. III. The area of any circular sector is half the rectangle contained by its arc and the radius of the circle. Let AOB be a sector. In the arc AB take any number of equidis- tant points A1,A 2,.....A n, and join AA1,A 1A2,.....A nB. Pro- duce AA1, and along it take parts A1A′ 2,A′ 2A′ 3,.....A ′ nB′ equal to A1A2,A 2A3,.....A nB respectively: so that AB′ is equal to the polygonal perimeter AA1A2.....A nB; then if the number of points A1,A 2, &c., be conceived to increase continually, the limit of AB′ is the arc AB.

[Chap. I.] OF THE MENSURATION OF THE CIRCLE. □ A1 A2 A′ 2 A3 A′ 3 A4 A′ 4 A5 A′ 5 O A B B′ B″T Now throughA draw the lineAT at right angles toOA, then as the number of points increases continually, the angle TAB′ shall diminish continually, and shall in time become less than any finite angle, and the limit of the position of AB′

shall be AB″ measured along AT , where AB″ is equal in length to the arc AB. JoinOA′ 1,OA′ 2,OA′ 3,.....OA ′ n and the triangles OA1A2,OA2A3, OA3A4,.....OA nB are equal, each to each, to OA1A′ 2, OA′ 2A′ 3, OA′ 3A′ 4,.....OA ′ nB, for the perpendiculars from O on the sides of the polygon are all equal to the perpendicular on AB′; therefore the variable triangle OAB′ is always equal to the variable polygon

[Chap. I.] TRIGONOMETRY. □ OAA1A2.....A nB; therefore their limits are equal. But the limit of the triangle OAB′ is OAB″ and the limit of the polygon is the sector OAB; therefore the sector AOB is equal to the triangle OAB″, which is half the rectangle OA, AB″, or half the rectangle contained by the radius and the arc. Hence the area of a circle = 1 2R× circumference = πR2 and the ratio of the circle to the square on its diameter is = π 4 . Prop. IV. Any line, whether curved or polygonal, which is convex throughout (that is, which can be cut by a straight line in only two points), is less than any line, curved or polygonal, which envelopes it from one extremity to the other ∗. For the enveloping line is obviously greater than the sum of any number of straight lines drawn as in Def. □, and therefore is greater than the limit of that sum, that is, than the length of the curve. Cor. Hence two straight lines, touching at its extremities any circular arc less than a semicircle, are together greater than the arc. Prop. V. If circles be inscribed in and described about two regular polygons of the same perimeter, the second of which has twice as many sides as the first, then (□) the radius of the circle inscribed in the second is an arithmetic mean between (i.e. is half the sum of) the radius of the circle inscribed in and the radius of the circle described about the first; and (□) the radius of the circle described about the second is a mean proportional between the radius of the circle inscribed in the second, and the radius of the circle described about the first. Let BB′ be a side of the first polygon, C the centre of the circle described about it. ∗This enunciation is taken from Legendre, Elements de Geometrie , 12 me ed. Liv. iv. Prop. ix., but the demonstration is different.

[Chap. I.] OF THE MENSURATION OF THE CIRCLE. □ FromC as centre with CB as radius describe the circle BB′E. DrawECA a diameter perpendicular toBB′ and therefore bisecting it in D. A B B′ C DE F G H JoinEB,EB′. Draw CF perpendicular to EB, and FGH perpen- dicular to EA. Then, because the angle BEB′ is half of BCB′, and FH is half of BB′, for FH bisects EB and EB′; therefore FH = the side of the second polygon, and FEH = the angle it subtends at the centre. ThereforeEF is the radius of the circle described about the second polygon, and EG the radius of the circle inscribed in it. AndCD,CB are the radii of the circles inscribed in and described about the first polygon.

[Chap. I.] TRIGONOMETRY. □ ButEG is half ofED, that is, half ofEC (orCB) andCD together, that is the radius of the circle inscribed in the second polygon is the arithmetic mean of the radii of the circles inscribed in and described about the first polygon. Again, because the triangles EFG , ECF are similar, EC :EF ::EF :EG, that is, the radius of the circle described about the second polygon is a mean proportional to that of the circle described about the first and that of the circle inscribed in the second. Cor. Hence the ratio of the circumference of a circle to its di- ameter (or π) can be calculated to any degree of accuracy. For letR,R′ be the radii of the circles

described about, and r,r′ of those inscribed in, the first and second polygon respectively, then r′ = R +r 2 ; R′ = √ r′·R. From these it will be easy to calculate successively the radii of circles inscribed in and described about isoperimetrical polygons of 2, 4, 8, 16, 32, &c. times the number of sides of a given regular polygon. Then, if the radii and perimeter of a regular polygon of any number of sides be known, by making it the first polygon of the series and calculating the radii for a sufficient number of succeeding polygons, we can calculate the value of π (the ratio of the circumference of a circle to its diameter) to any degree of accuracy. For since the perimeter of each polygon will lie between the circumference of its inscribed and circumscribed circles if R and r be the radii for any polygon of the series, we shall have 2 πR greater, and 2 πr less than p, the common perimeter of all the polygons. Therefore π is intermediate to p 2R and

[Chap. I.] OF THE MENSURATION OF THE CIRCLE. □ p 2r, and, by doubling the number of sides of the polygon sufficiently, R andr can be made to differ as little as we please, and thereforeπ can be calculated as accurately as desired. The calculation is not very laborious. Thus, if we begin from a square, each side of which is the unit, we have r1 = 0.5 and R1 = √ .5 = 0.7071067812. Then r2 = 0.5 + 0.7071067812 2 = 0.6035533906, and R2 = √ .7071067812×.6035533906 = 0.6532814824. In like manner the radii of circles inscribed in and described about polygons of 16, 32, 64, 128, &c. sides with the same perimeter (viz. 4) are successively found by alternately taking arithmetic and geometric means. Stopping at the polygon of 1024 sides, it appears that 2000000 636621 <π < 2000000 636617 , i.e. 3.14158<π < 3.14160. It may however be shewn (see Appendix) that, when the difference between R and r is small, 1 3(r + 2R) is a very near approximation to the limit of both radii, and that thereforeπ may be taken = 1 2p 1 3(r + 2R) with great accuracy.

[Chap. I.] TRIGONOMETRY. □□ No. of sides of the Polygon. Radius of Inscribed Circle = r. Radius of Circumscribing Circle = R. 4 .5000000000 .7071067812 8 .6035533906 .6532814824 16 .6284174365 .6407288619 32 .6345731492 .6376435773 64 .6361083633 .6368755077 128 .6364919355 .6366836927 256 .6365878141 .6366357516 512 .6366117828 .6366237671 1024 .6366177750 .6366207710 &c. &c. &c. Taking the radii for 1024 sides r + 2R 3 = 1 3 { .6366177750 1.2732415420 } = 1.9098593170 3 =.6366197723, which will be found to be ten decimals of the radius of the circle in- scribed in a polygon of 262144 and every greater number of sides if the table be continued. Thus we may take π = 20000000000 6366197723 , or = 3 .141592654. By the method of “continued fractions” it will be found that 22 7 and 355 113 are nearer approximations to the value of π than any simpler fractions.

[Chap. I.] OF THE MENSURATION OF THE CIRCLE. □□ Of these 22 7 (= 3.14) is the approximation discovered by Archimedes (killed, it is said, at the siege of Syracuse, b.c. □□□); and the approxi- mation 355 113 (= 3.14159) was given by Adrian Metius of Alkmaer (died a.d. □□□□)∗. ∗This simple and elegant elementary method of approximating to π is taken from Leslie’s Geometry, v. 20; compare Legendre, Geometrie, iv. 14 and 16.

CHAPTER II. OF THE AREA OF A TRIANGLE AND OF THE INSCRIBED CIRCLE. Prop. I. A triangle is equal to the rectangle contained by its semi- perimeter and the radius of the inscribed circle. Let ABC be the triangle. Bisect the angles by the lines AO, BO, CO, meeting (Euclid iv. □) in O, the centre of the inscribed circle. Then the triangle ABC is made up of the triangles BOC , COA, AOB, each of which stands on one of the sides, as base, with its altitude equal to the radius of the inscribed circle. Therefore the whole triangle ABC is equal to a triangle having the sum of the three sides (or the perimeter) for base and the radius of the inscribed circle for altitude; or to the rectangle having the semi-perimeter for base and the radius of the inscribed circle for altitude. Scholium. The two tangents from each angle to the inscribed circle are equal: hence, if three tangents, one from each angle, be taken, their sum is the semi-perimeter, and therefore a tangent from one of the angles, together with the side opposite that angle, is equal to the semi-perimeter. Let the sides opposite the anglesA,B,C be represented numerically bya,b,c; the semi-perimeter bys, and the radius of the inscribed circle byr. Then, numerically, the Area = rs. And Ab =Ac =s−a, Bc =Ba =s−b, Ca =Cb =s−c. Def. Let two of the sides of the triangleABC be produced, and a circle described touching the two produced sides and the third side. □□

 The circle is said to be excribed∗ on the third side. Prop. II. A triangle is equal to the rectangle contained by the ra- dius of the circle excribed on one of its sides and the tangent from the opposite angle to the inscribed circle. A B C a b c a′ b′ c′ O O′ LetABC be the triangle. Bisect the angle A and the exterior angles at B and C by the lines AO′, BO′, CO′, which will meet in the centre ∗This word is often spelled “ escribed” improperly. The Latin word is exscribo, but the English usage is to elide the s in such cases, as expect from exspecto, expatiate from exspatior, extinguish from exstinguo. No one ever proposed to emend these words into espect, espatiate, and estinguish. Why then escribe?

 of the excribed circle ∗. Draw perpendiculars O′a′, Ob′, O′c′ on the three sides. Then these perpendiculars are all equal and each of them is a radius of the excribed circle. Also the two tangents from each angle to the excribed circle are equal; and therefore Ba′ is equal toBc′, Ca′ to Cb′, and Ab′ to Ac′. Hence Ab′ and Ac′ are together equal to the perimeter of the triangle and each of them to the semi-perimeter. But because the two triangles AcO and Ac′O′ are similar therefore Ac :Oc ::Ac′ :O′c′ and (Euclid vi. □□) the rectangle Ac, O′c′ is equal to the rectangle Oc, Ac′, which by Prop. I. is the area of the triangle. Numerically, if the radius of the circle excribed on the side BC be represented by α, this may be written (s−a)α =rs = the area. Prop. III†. A triangle is a mean proportional to the rectangle con- tained by the semi-perimeter and its excess over one of the sides, and the rectangle contained by the excess of the semi-perimeter over each of the other sides. With the same figure as before the right-angled triangles BOc and BO′c′ have the angles BOc and OBc equal to the angles O′Bc′ and BO′c′ each to each. Therefore these triangles are similar and (Eu- clid vi. □) Bc :cO ::c′O′ ::c′B. Now, on the first and second of these lines let rectangles of alti- tude Ac′ be constructed, and on the third and fourth rectangles of altitude Ac. Then (Euclid vi. □) Ac′,Bc :Ac′,cO ::Ac,c′O′ :Ac,c′B. But the rectangle Ac′, cO is equal to the triangle ABC by Prop. I. and the rectangle

Ac,c′O′ is equal to ABC by Prop. II. ∗The proof of this is left to the reader, or he may consult Thomson'sEuclid, iv. 4. †This most useful proposition was known to the Greeks of Alexandria, and by them communicated to the Arabians, but seems to have "been reinvented in Europe about the latter part of the □□th century." Leslie's Geometry, □□□□, v. □□, where the above demonstration (nearly) will be found.

[Chap. II.] OF THE AREA OF A TRIANGLE □□ Therefore the triangle is a mean proportional between the rectan- gles Ac′, Bc and Ac, c′B, that is between the rectangle contained by the semi-perimeter and its excess over the side CA, and the rectangle contained by its excess over the sides BC and AB respectively. Writing this numerically, and supposing the area to be represented in square units by the number Δ, it becomes s(s−b) : Δ :: Δ : (s−a)(s−c) or Δ2 =s(s−a)(s−b)(s−c), whence the area can be calculated in square units when the lengths of the sides are given numerically in units. Also by Prop. II. r2 = (s−a)(s−b)(s−c) s .

CHAPTER III. OF SYMBOLS OF QUANTITY. Angles not limited in magnitude. In Euclid an angle is not defined as a magnitude but as the inclination of two lines, which never exceeds two right angles: and in most of the propositions in Euclid it is not necessary to treat an angle otherwise than as a change of direction of a line. But where an angle is treated as a magnitude (notably in Euclid vi.□□ and consequently in iii. □, on which it depends) any multiple whatever of an angle is termed an angle. So also in Trigonometry, where angular magnitude in general is treated numerically, it is desirable to use the term angle for the sum of a number of angles, which may be greater than two or than any number of right angles. In the same way an arc of a circle may be greater than a circumference or any number of circumferences. Negative quantities. Again, when magnitudes are represented nu- merically by algebraic symbols, the values of which are defined but not specified, it is often desirable to express a difference without limiting the generality of the expression by stating which of the symbols stands for the greater number. For instance, if a distance a miles be measured from a fixed pointO along (or parallel to) a given line in a standard di- rection, say east, toA; and a line AB be cut off from OA by measuring from A in the opposite direction, or westward, a distance b miles, the distanceOB may be said to be = (a−b) miles east ofO not only in the case where a>b , but also when a<b , if it be agreed to interpret the result as meaning a−b east of O (or in the standard direction) if a−b is a + number, and b−a miles west ofO (or in the contrary direction), when a−b is a− number. Then the standard direction may be called the + (or positive) di- rection, and the contrary direction the − (or negative) direction. □□

[Chap. III.] OF SYMBOLS OF QUANTITY. □□ In the same way, ifA bea feet aboveO andB beb feet belowA,B is a−b feet above O ifa>b , and b−a feet below O ifa<b . To express the result by one formula we may say that B is a−b feet above O in both cases, if we interpret the + sign as meaning upwards from O and the − sign as meaning downwards (or in the contrary direction) from O. Thus, in the case of lines measured along (or parallel to) a specified line from a given point (or origin) the sign + is conveniently prefixed (or understood) before lines measured in a standard direction and the sign− before those measured in the contrary direction. Again, with reference to angles, if the hand of a going clock be put back through an angle θ, then,

after the time during which the hand moves through an angle φ, the hand will make an angle θ−φ with its present position, the angle being + and measured in the opposite way to that in which the hand of the clock moves, if θ > φ; or− and measured in the contrary direction, if θ <φ. In what follows, an angle will be considered as if produced by the revolution of a radius of a circle, the direction of revolution from an initiatory position being considered as − or + according as it takes place in the direction of the motion of the hand of a clock or the reverse.

CHAPTER IV. OF THE UNIT OF ANGULAR MAGNITUDE. In order to treat angular magnitude numerically it is necessary to use some fixed angle as a standard of comparison, by reference to which the magnitudes of angles under consideration may be denoted. The angle of easiest construction is the angle of an equilateral tri- angle, which is also two-thirds of a right angle. For the purpose of expressing simply fractions of the standard, the sexagesimal division (or division into 60ths) of the standard is proba- bly the most convenient (because the third, fourth, fifth, sixth, tenth, twelfth, fifteenth, twentieth and thirtieth are all exact numbers of six- tieths). For such reasons perhaps the sexagesimal scale, which has prevailed since the time of Ptolemy∗, was originally adopted. It is still employed, and we have the following Notation for angles in aliquot parts of a right angle : — The 90th part of a right angle (or the 60th of the angle of an equi- lateral triangle) is called a degree. One degree is denoted by 1◦; so that a right angle is 90 ◦; the angle of an equilateral triangle is 60◦. The 60th part of a degree is a minute, denoted by 1 ′, □ 1◦ = 60′. The 60th part of a minute is a second, denoted by 1 ″, □ 1′ = 60″. Fractions of a second are now usually denoted by decimals, but in older books, as for instance in Newton’s Principia, the sexagesimal division is carried farther, so that 1″ = 60‴, 1‴ = 60iv, 1iv = 60v. ∗See article “Arithmetic” in the Encyclopædia Metropolitana, p. □□□,§□□. □□

[Chap. IV.] UNIT OF ANGULAR MAGNITUDE □□ This notation is used for many practical purposes ∗. Circular Measure. In formulæ, involving explicitly the numerical value of an angle, it is more suitable and it is usual to represent the angle by its ratio to the angle subtended at the centre of a circle by an arc equal to the radius. This angle (Chap. I. Prop. I. Cor. □) is invariable, that is, is the same whatever radius be taken, and can therefore be used with propriety for a standard of comparison. It is called the unit of circular measure, and the ratio of any angle to this unit is called the circular measure of the angle. The circular measure of an angle is also the ratio of the arc sub- tending the angle at the centre of any circle to the radius. For let AB be the arc subtending the angle AOB, of which the circular measure O A A′ B B′ C is θ, at the centre of the circle of which the radius OA = R. And let AOC be subtended by the arc AC = R. Then AOC is the unit; and (Euclid vi. □□) AOB :AOC ::AB :AC, or θ = AB AC = AB R , and the arc AB =Rθ. ∗The centesimal division of the right angle into 100 grades, &c. proposed at the French Revolution, though adopted in the M ′ ecanique C ′ elesteof Laplace, has been abandoned even in France.

[Chap. IV.] TRIGONOMETRY. □□ The ratio of the arc to the radius is obviously the same whatever radius be taken. For if A′B′ be the arc subtending the same angle at the centre of the circle of radiusOA′ =R′, we haveAB :A′B′ ::R :R′, or AB R = A′B′ R′ . The circular measure of a right angle = 1 4 circumference R = π 2. And the number

of degrees in the unit of circular measure = 180◦ π . Complementary angles. Two angles are said to be complements, each of the other, when their sum is a right angle. Hence the comple- ment of an angle A◦ contains 90◦−A◦, and the circular measure of the complement of θ is π 2−θ. If A◦ > 90◦, or θ >π 2 , the complement is− and to be measured in the negative direction. Supplementary angles. Two angles are supplements, each of the other, when their sum is two right angles. Hence the supplement of A◦ is 180◦−A◦; and the circular measure of the supplement of θ is π−θ. IfA◦ > 180◦, or θ >π, the supplement is− and to be measured in the negative direction.

CHAPTER V. CIRCULAR FUNCTIONS, OR TRIGONOMETRICAL RATIOS. Function. When one magnitude or ratio is so connected with an- other that the former changes with the latter, but is determinable for any given value of the latter, the former is said to be a function of the latter. Hence certain ratios, which depend on the value of the angle, or its circular measure, are called circular functions. A A′ BB′ B″ B‴ C C ′ DD′ E O LetACA′C′ be any circle,O its centre,A′OA,COC′ two diameters at right angles dividing the circle into four quadrants. □□

[Chap. V.] TRIGONOMETRY. □□ Let OA, OC be the + directions for lines measured from O along these lines respectively. Let OA be taken as the initial line for angles, andABC the + direction for angles measured from OA. Let R be the radius of the circle, and θ (AB R) be the circular measure of AOB. □. Sin θ. The sine of the angle AOB is the ratio to the radius of the perpendicular from the end of the arc subtending the angle on the initial line OA, □ sinθ = BD R . If the position of B be above the line AOA′, sin θ is +; if below, sinθ is−. □. Cos θ. The cosine of an angle is the sine of its complement, or cosθ = sin (π 2−θ) . Conversely sinθ = cos (π 2−θ) . The angle AOB being θ, and COA = π 2 , COB = π 2−θ; therefore cosθ = sinCOB = BE R = OD R . Hence, ifOD is measured towardsA, or ifB lie to that side ofCOC′, cosθ is +; if OD is measured towards A′, or if B lies on the same side of COC′ as A′, cosθ is−. Variation of the Sine and Cosine. We may examine how the values of these two functions change with the variation of the angle.

[Chap. V.] CIRCULAR FUNCTIONS □□ When AB is small, BD is small and +; OD = the radius nearly and +. As θ increases from 0 to π 2 , BD and OD remain both +, but BD increases from 0 to R, and OD decreases from R to 0; therefore sinθ increases from + 0 to + 1; cosθ decreases from + 1 to 0 . Asθ increases from π 2 toπ,BD remains +, but diminishes from +R to 0; also OD becomes−, and−OD increases from 0 to +R; therefore + sinθ decreases from + 1 to 0; − cosθ increases from 0 to + 1 . As θ increases from π to 3π 2 , BD becomes−, and−BD increases from 0 to +R; also OD is−, and−OD decreases to 0; so that − sinθ increases from 0 to + 1; − cosθ decreases from + 1 to 0 . As θ increases from 3π 2 to 2π, −BD decreases to 0; and + OD increases to +R, so that − sinθ diminishes from + 1 to 0; + cosθ increases from 0 to + 1 . After this the values from 2 π to 4π are the same as from 0 to 2 π: and if m be any whole number sinθ = sin(2mπ +θ); cosθ = cos(2mπ +θ).

[Chap. V.] TRIGONOMETRY. □□ If equal angles, + θ and−θ, be measured in opposite directions from OA, and the other ends of the arcs subtending them be joined by the straight line BB‴, AOA will bisect BB‴ at right angles in the point D.

Therefore the sines of these two angles will be equal but of opposite sign, and their cosines will be equal in sign as well as magnitude; or sin(−θ) =− sinθ; cos(−θ) = + cosθ. IfA′B′,A′B″ be arcs, each = AB =Rθ, but measured from A′; the circular measure of the angles subtended by AB′,AB″ are respectively π−θ, π +θ; and the perpendicular B′D′ = +BD, B″D =−BD, and OD′ =−OD. Consequently sin(π−θ) = sinθ, cos(π−θ) =− cosθ; sin(π +θ) =− sinθ, cos(π +θ) =− cosθ. And, since both sine and cosine remain unchanged when the angle is increased by a multiple of 2π (say by 2mπ), we have sin(2mπ +θ) = + sinθ; sin(−θ) = sin(2mπ−θ) =− sinθ; sin(π−θ) = sin ((2m + 1)·π−θ) = + sinθ; sin(π +θ) = sin ((2m + 1)·π +θ) =− sinθ. And cos(2mπ +θ) = + cosθ; cos(−θ) = cos(2mπ−θ) = + cosθ; cos(π−θ) = cos ((2m + 1)·π−θ) =− cosθ; cos(π +θ) = cos ((2m + 1)·π +θ) =− cosθ.

[Chap. V.] CIRCULAR FUNCTIONS □□ By these the sine and cosine of an angle of any magnitude can be obtained from the sine or cosine of an acute angle. It should be observed that, while the number θ continuously in- creases, the numbers sinθ, cosθ pass through a series of values between +1 and−1, and return to the same values again for every increase of 2π in the value of θ. They are therefore said to be periodic functions ofθ, of which the period is 2π. Since the perpendicular from the centre of a circle on any chord bisects it at right angles, the ratio of the chord to the radius is twice the sine of half the angle subtended by the chord. Hence, if we can calculate the ratio to the radius of the side of an inscribed polygon, the sine of half the angle subtended by the side is at once known. For instance, the side of a square inscribed in a circle = R √ 2, and it subtends an angle of π 2 or 90◦. Therefore sinπ 4 = sin 45◦ = 1 2 √ 2 = cos 45◦ = cosπ 4. The side of the hexagon inscribed is = R, and it subtends π 3 or 60◦. Therefore sinπ 6 = sin 30◦ = 1 2 = cos 60◦ = cosπ 3. The side of the inscribed equilateral triangle = R √ 3, and it sub- tends 2π 3 or 120◦. Hence sinπ 3 = sin 60◦ = 1 2 √ 3 = cos 30◦ = cosπ 6.

[Chap. V.] TRIGONOMETRY. □□ The side of the regular decagon inscribed is (Euc. iv.□□, and ii.□□)√ 5− 1 2 R, and it subtends an angle π 5 or 36◦. Hence sin π 10 = sin 18◦ = √ 5− 1 4 = cos 4π 10 = cos 72◦. □. Tan θ. The tangent of an angle is the ratio to the radius of the part of the line, touching at the initial end the arc subtending the AA′ BB′ B″ B‴ C F F ′ G G′ H H ′ T T ′ O angle, intercepted between the point of contact and the radius produced through the other end of that arc. In the diagram tanθ = AF R and AF =R tanθ.

[Chap. V.] CIRCULAR FUNCTIONS □□ IfAF be measured towards T , it is to be considered +, and −, if in the contrary direction towards T′. □. Cot θ. The cotangent of an angle is the tangent of the comple- ment of the angle. Whence cotθ = tan (π 2−θ) and tan θ = cot (π 2−θ) . The cotangent may hence be geometrically defined as follows. Let OC be drawn perpendicular to OA. Then COB is the complement of AOB, or COB = π 2−θ. If AOB be greater than AOC, or θ > π 2 , COB will be measured from CO in the other direction, and π 2−θ will be− to correspond. AA′ BB′ B″ B‴ C F F ′ G G′ H H ′ T T ′ O

[Chap. V.] TRIGONOMETRY. □□ Draw H′CH touching the arc subtending the complement of θ at the point C, which is always an extremity of that arc, and through the other extremity of the arc produce the radius to meet it; the tangent of the

complement, or cotangent, is the ratio to the radius of the part of HCH′ intercepted. In the diagram cotθ = CG R and CG =R cotθ. IfCG is measured in the direction fromC toH′ the cotangent is−. Variation of the Tangent and Cotangent. In the first quadrant, when the angle is indefinitely small, the tangent is indefinitely small, and the cotangent indefinitely great; since AF diminishes, and CG increases without limit, as AOB decreases. Hence it is usually said that tan 0 = 0, and cot 0 =∞. As the angle increases to 90◦,AF increases andCG decreases, both drawn in the + direction; so that tan θ increases and cot θ decreases, as θ increases from 0 to π 2 , and when θ = π 2 , tanθ =∞, cotθ = 0. In the second quadrant, where θ increases from π 2 to π, AF′ is measured in the− direction AT′, and CG′ in the− direction CH′ for an angle AOB′: so that tan θ and cotθ are both−. Also, AF′ decreases from an indefinitely great distance in the di- rection AT′, when θ barely exceeds π 2 , to 0 when θ = π; while CG′ increases from 0 to an indefinitely great distance in the direction CH′. Hence, as θ increases from π 2 to π,− tanθ decreases from∞ to 0 and − cotθ increases from 0 to∞. In the third quadrant, or when θ increases from π to 3π 2 , the lines AF and CG are again measured in the positive direction, and the

[Chap. V.] CIRCULAR FUNCTIONS □□ tangent and cotangent are therefore both +, and vary in magnitude as in the first quadrant. In the fourth quadrant, where θ is greater than 3π 2 , and less than 2π, the lines are again drawn in the − direction, and the tangent and cotangent vary just as in the second quadrant. If two equal angles, as AOB and AOB‴, or AOB′ and AOB″, be measured in opposite directions from OA, it is obvious that their tangents and cotangents are equal in magnitude, but opposite in sign; or tan(−θ) =− tanθ; and cot(−θ) =− cotθ. Also if A′OB′ and AOB are equal angles, so that AOB′ and AOB are supplementary, their tangents and cotangents are equal, but of op- posite signs; that is tan(π−θ) =− tanθ; cot(π−θ) =− cotθ. Ifm be any whole number, it is clear that the angle 2mπ +θ begins and terminates at the same point as the angle θ. Therefore, tan(2mπ +θ) = tanθ; cot(2 mπ +θ) = cotθ; tan(2mπ−θ) =− tanθ; cot(2 mπ−θ) =− cotθ; tan ((2m + 1)·π−θ) =− tanθ; cot ((2m + 1)·π−θ) =− cotθ; tan ((2m + 1)·π +θ) = tanθ; cot ((2m + 1)·π +θ) = cotθ. Whence the tangent and cotangent of any angle can be found from those of an acute angle. □. Sec θ. The secant of an angle is the ratio to the radius of the initial radius produced to meet the tangent at the other end of the arc subtending the angle.

[Chap. V.] TRIGONOMETRY. □□ AA′ BB′ B″ B‴ C K K ′ SS′ O □. Cosecθ. The cosecant is the secant of the complement. In the diagram, AOB being the angle, and COB the complement, KS the tangent at B; secθ = OS R ; cosec θ = OK R ; OS =R secθ; OK =R cosecθ. Then sec 0 = 1; and the secant increases till sec π 2 =∞. When θ is between π 2 and 3π 2 , sec θ is−, and − secθ decreases from ∞ (when θ = π 2) till − secπ = 1, and then increases again till − sec 3π 2 =∞;

[Chap. V.] CIRCULAR FUNCTIONS □□ after which, while θ increases from 3π 2 to 2π, + secθ decreases from∞ to 1. Similarly cosecθ is + in the first and second quadrants diminishing from cosec 0 =∞ till cosec π 2 = +1 and then increasing till cosec π = +∞. In the third and fourth quadrants cosec θ is−, and − cosecθ de- creases from∞ at the beginning of the third quadrant till− cosec 3π 2 = 1, and then increases till − cosec 2π =∞. □. Versinθ. The versed sine of an angle is the ratio to the

radius of A A′ BB′ B″ B‴ C C ′ DD′ E O the part of the initial radius (produced if necessary) which is intercepted

[Chap. V.] TRIGONOMETRY. □□ between its extremity A and the perpendicular from the other end B of the arc subtending the angle. In the diagram, we have versinθ = AD R ; and AD =R versinθ. The line AD is always drawn in one direction and versin θ is al- ways +. As θ increases from 0 to π, versinθ increases from 0 to 2; and as θ increases from π to 2π, versinθ diminishes from 2 to 0. The seven ratios defined above are altogether independent of the size of the circle described, and depend only on the angle. They are therefore functions of the circular measure θ, remaining the same for the same value of θ whatever radius be taken, but changing with the value of θ; and they are numbers calculable from the number θ. The lines, to which their ratios correspond, depend partly on the value of θ and partly on the radius, and are expressed by the radius multiplied by the corresponding circular function. Thus if the angle AOB =θ in the diagrams of this Chapter, BD =OB sinθ =OD tanθ, EB =OB cosθ =EO cotθ, AF =OA tanθ =OF sinθ, CG =OC cotθ =OG cosθ, OS =OB secθ, OK =OB cosecθ. Relations between the circular functions of the same angle. If the angleAOB and the radius of the circle be the same in these diagrams, the triangles BOD, FAO , SOB are right-angled and similar. Hence

[Chap. V.] CIRCULAR FUNCTIONS □□ expressing their sides in terms of the radius and the circular function corresponding to each, we have (Euc. i. □□) R2 =R2 sin2θ +R2 cos2θ, or sin 2θ + cos2θ = 1. (□) R2 sec2θ =R2 +R2 tan2θ, or sec 2θ = 1 + tan2θ. (□) And putting π 2−θ for θ in this cosec2θ = 1 + cot2θ. (□) Also (Euc. vi. □) R tanθ :R ::R sinθ :R cosθ, or tan θ = sinθ cosθ (□) R secθ :R ::R :R cosθ, or sec θ = 1 cosθ. (□) From which putting π 2−θ for θ, cotanθ = cosθ sinθ = 1 tanθ (□) cosecθ = 1 sinθ (□) And R versinθ =R−R cosθ whence versin θ = 1− cosθ. (□) From these equations all the functions can be found, when one has been given. For instance, from the values of the sine and cosine given

[Chap. V.] TRIGONOMETRY. □□ above (p. □□) we get tan 45◦ = cot 45◦ = 1, sec 45◦ = cosec 45◦ = √ 2, tan 30◦ = cotan 60◦ = 1√ 3 = 1 3 √ 3, sec 30◦ = cosec 60◦ = 2√ 3 = 2 3 √ 3, cot 30◦ = tan 60◦ = √ 3 = √ 3, cosec 30◦ = sec 60◦ = 2, cos 18◦ = 1 4 √ 10 + 2 √ 5, &c.

CHAPTER VI. OF LOGARITHMIC TABLES. Tables have been formed of the values of the circular functions of angles at intervals of 1 ′, or in more minute tables at intervals of 10 ″, from 0◦ to 45◦. It is unnecessary to calculate those of greater angles, as the angles between 45◦ and 90◦ are complements of the angles between 0◦ and 45◦. Methods of calculation are given in Higher Trigonometry. Such tables are called Tables of Natural Sines, Cosines, &c. For prac- tical calculations they are superseded by logarithmic tables, the use of which may be shortly explained. Logarithms of ordinary numbers may be defined to be numbers, so calculated from the ordinary numbers, that the sum of the logarithms of two numbers is the logarithm of their product. Thus, if m and n be two numbers, logm + logn = log(m×n). (□) Now log(m÷n) + logn = logn(m÷n) = logm; □ logm− logn = log(m÷n). (□) From (□) it follows, putting m =n, that log 1 = 0, (□) and that the logarithm of a proper fraction is −. Also from (□) logn2 = 2 logn, logn3 = 3 logn, &c. □□

[Chap. VI.] TRIGONOMETRY. □□ and, generally, lognp =p logn, if p is an integer, and therefore q logn p q =p logn, or log n p q = p q× logn. So that whether x is an integer, or a fraction, lognx =x logn. (□) Ifa = the number of which the logarithm is 1,a is said to be the base of the system of logarithms: then denoting logarithms in the system of which the base is a by loga, (□) becomes logan =x, when ax =n. (□) Common logarithms are those for which the base is 10. Hence in (□) and (□) denoting common logarithms by log; log 1 = 0; log 10 = 1; log 100 = 2; log 1000 = 3; and so on. Equation (□) shews that the logarithm increases with the number: therefore it appears that the common logarithm of a number of one integral digit is a proper fraction; that of a number of 2 digits is 1 + a fraction; of 3 digits 2 + a fraction, and so on. The integral part, or characteristic, of the logarithm, therefore, is one less than the number of integral digits and is known by inspection, so that it is usual in the tables to give only the decimal part of the logarithm, or the mantissa. Since also by (□) and (□), log(10m×n) =m + logn, and log(n÷ 10m) =−m + logn,

[Chap. VI.] OF LOGARITHMIC TABLES. □□ it appears that multiplication or division by any power of 10 will affect only the characteristic of the logarithm, and that the decimal part of the logarithm, or mantissa, will be the same for numbers consisting of the same succession of digits followed or preceded by any number of ciphers, and with a decimal point occurring between any two digits. It will be seen that all the arithmetical operations for calculation are simplified by the use of logarithmic tables. Thus, with tables, the result of multiplication is calculated by adding logarithms, of division by subtracting logarithms, of raising to a power by multiplying a logarithm by a number, of extracting a root by dividing a logarithm by a number. Calculations are thus so much abbreviated∗ that it is very desirable in establishing formulæ for the calculation of some of the sides and angles of triangles from given values of others, that the formulæ should be adapted to logarithmic calculation, as is done in the next chapter. It is also necessary to have logarithmic tables of the Trigonometrical ratios. Since the sine of an angle is always less than 1, the common loga- rithm of the sine is a − number. The same is true of the cosine, of the tangent of an angle < 45◦, &c. To avoid the use of negative numbers, the tabular logarithms of the circular functions are the common loga- rithms of these ratios increased by 10: which must be remembered in using the tables in calculations. Where, as is generally the case in practice, the logarithm of a given number, or, conversely, the number for a given logarithm is not actually to be found in the tables, these can be found approximately from the ∗Laplace has remarked with reference to the value of the invention, that “by reducing to a few hours what otherwise is the labour of several months, it doubles, so to speak, the life of the astronomer, and spares him the errors and annoyance inseparable from long calculations.”

[Chap. VI.] TRIGONOMETRY. □□ principle that, for considerable numbers of small difference, the differ- ence of two numbers is nearly proportional to the difference of their logarithms. Whence the difference between a number and the nearest number in the tables being known, the difference to be added to the nearest logarithm in the tables can be approximately found, and the converse. The same method applies also to the tabular logarithms of the circu- lar functions, it being generally approximately true that the difference of two tabular logarithms of a circular function is

proportional to the difference of the corresponding angles, when that difference is small, and the angles are neither very small nor very near 90 ◦∗. ∗Logarithms were invented by John Napier, who was born in □□□□ and died □th April, □□□□. He was descended from and claimed to represent the elder branch of the Earls of Lennox. From him are descended the Baronet of Milliken and the Lord Napier. But the inventor of Logarithms was not a Peer, and should not be styled Baron Napier as is often done. The mistake arises from his having been “Baron” in the sense of proprietor of the Barony of Merchistoun, near Edinburgh. In a paper dated □□rd April, □□□□, at Gartness on the Endrick (where were his Lennox Estates and where it is a local tradition that he calculated his Canon of Logarithms), he signs himself “Jhone Neper Fear of Merchiston.” The word fear is the Celtic equivalent of Baron. His “Canon,” or table of Logarithms, was published in □□□□.

CHAPTER VII. SOLUTION OF TRIANGLES. A triangle is said to be solved when the sides and angles are calculated from the data. Right-Angled Triangles. Let ABC be a triangle. Let C = 90◦; and let BC =a, CA =b, AB =c. A B C Then if the side AB be taken as radius, the ratio of BC to AB is the sine of A; and of AC to AB is its cosine; or a =c sinA; b =c cosA. Similarly b =c sinB; a =c cosB. If AC be taken as radius, since CB touches the arc described, the ratio of CB to CA is the tangent, and of AB to CA is secant of A; so that a =b tanA; c =b secA. And similarly b =a tanB; c =a secB. From these equations, it is easy logarithmically to solve a right- angled triangle when a side and an angle, or when two sides are given. □□

[Chap. VII.] TRIGONOMETRY. □□ Given c and A, we have B = 90◦−A, loga = logc + tab log sinA− 10, logb = logc + tab log cosA− 10; whence B, a, b are found. Given a and A, we have B = 90◦−A, logb = loga + tab log tanB− 10, logc = loga + tab log secB− 10; whence B, c, b are found. Given c and a, we have tab log sinA = 10 + loga− logc, B = 90◦−A, logb = logc + tab log cosA− 10; whence A, B, b are found. Triangles not Right-Angled. Case I. Let ABC be an acute-angled triangle, ABC′ a triangle D A B C D A B C ′ having an obtuse angle atC′. In both triangles draw AD perpendicular

[Chap. VII.] SOLUTION OF TRIANGLES. □□ to BC produced if necessary. Then in both triangles, if AB be taken as radius, AD =AB sinB =c sinB; and if CA be taken as radius, AD =CA sinC =b sinC. Hence b sinC =c sinB, or b sinB = c sinC. If the perpendicular had been drawn from B toCA we should sim- ilarly have got a sinA = c sinC , so that a sinA = b sinB = c sinC , or the sides are proportional to the sines of the opposite angles . This is true of all triangles. Hence we can logarithmically solve a triangle when a side and the opposite angle are given, and likewise another side or another angle. Given C and c, and b. Tab log sinB = (tab log sinC− logc) + logb, a logarithmic equation for the determination of B. There is but one acute angle which can have its sine of a particular value: so that, if we know B to be acute, there is but one value of B to be taken. And if b<c the angle B must be acute; and we find it by means of the equation.

[Chap. VII.] TRIGONOMETRY. □□ Then A = 180◦− (B +C), and loga = tab log sinA− (tab log sinC− logc), so that the triangle is solved. If however b > c, we do not know that B is acute, and as the sine of the supplement of an acute angle is the same

as the sine of the angle, there are two values of B, the one less than 90 ◦, the other its supplement, which satisfy the above logarithmic relation. That is, there is an ambiguity as to the value of B. There will always be this ambiguity in determining the angle of a triangle from its sine, unless there be something to point out whether the angle is acute or obtuse: and therefore it is generally inconvenient to use a method involving the determination of the angle from its sine. But in this instance there is a real geometrical ambiguity. For if we construct a triangle with its angle ACB = C, its side CA = b, and the side opposite C = c, it is evident that, if c be < b, there are two triangles AB1C and AB2C corresponding to the data. The angle opposite CA in the one of these is acute = B1; in the other it is obtuse = B2. But since AB1 = AB2, B2 = 180◦−B1. In this case therefore, where two sides and the angle A B1 B2 C opposite the lesser are given , which is called the ambiguous case, there are two triangles. The acute angle B1 can be found from the formula tab log sinB1 = logb + (tab log sinC− logc),

[Chap. VII.] SOLUTION OF TRIANGLES. □□ and then B2 = 180◦−B. If A1 and A2 are the corresponding values of the third angle a1, a2 of the third side, A1 = 180◦− (B1 +C); A2 = 180◦− (B2 +C), and loga1 = tab log sinA1− (tab log sinC− logc), loga2 = tab log sinA2− (tab log sinC− logc). Case II. Let ABC be a triangle: and let CB be greater than CA. A B CD E F From C as centre with CB as radius, describe the circle DBE . Produce BA to meet its circumference in F , and CA both ways to meet it in E and D.

[Chap. VII.] TRIGONOMETRY. □□ Join FC , BD, BE, BF . Then DBE is a right angle; and DCB =A +B = 2BED ; also A =ECF +CFA =ECF +B, and ECF =A−B = 2FBE. So that BED = 1 2(A +B); FBE = 1 2(A−B). And BDA is the complement of BED , ABD of FBE . Now in the triangle DBA, DA sinABD = BA sinBDA, or a +b cos 1 2(A−B) = c cos 1 2(A +B) . And in the triangle BAE, EA sinFBE = AB sinBED, or a−b sin 1 2(A−B) = c sin 1 2(A +B) . By combining these and remembering that tan θ = sinθ cosθ we get tan 1 2(A−B) tan 1 2(A +B) = a−b a +b.

[Chap. VII.] SOLUTION OF TRIANGLES. □□ These relations give the simplest logarithmic solution of a triangle, when two sides and the contained angle are given∗. Given a, b, and C, we have A +B = 180◦−C, so that 1 2(A +B) is known. Then tab log tan1 2(A−B) = tab log tan1 2(A +B) + log(a−b)− log(a +b), which determines the acute angle 1 2(A−B). Whence A = 1 2(A +B) + 1 2(A−B) and B = 1 2(A +B)− 1 2(A−B) are found. The third side c could now be found from the relation log c = tab log sinC− (tab log sinA− loga), but it is shorter in practice to determine c either from logc = tab log cos1 2(A +B)− tab log cos1 2(A−B) + log(a +b), ∗These formulæ are given in p.□□ of Thacker’s Miscellany,□□□□. Their applica- tion to the solution of triangles was pointed out by Professor Wallace, of Edinburgh, in the Transactions, R.S.E., □□□□.

[Chap. VII.] TRIGONOMETRY. □□ or from logc = tab log sin1 2(A +B)− tab log sin1 2(A−B) + log(a−b), Thus when two unequal sides and the angle between them are given, the other side and other angles are found. If an angle and two equal sides containing it are given, the remaining angles are each the complement of half the given angle, and the third side can be found as in Case I. Case III. Let ABC be the

triangle, O the centre of the inscribed circle, so that AO,BO,CO bisect the angles, and the perpendiculars O A B Ca bc Oa,Ob,Oc are each of them a radius of the inscribed circle. Let this radius = r, let the sum of the sides a +b +c = 2s; then (Chapter ii.) Ab =Ac =s−a, Bc =Ba =s−b, Ca =Cb =s−c.

[Chap. VII.] SOLUTION OF TRIANGLES. □□ And in the right-angled triangles AOb, BOc, COa, we have r = (s−a) tan 1 2A = (s−b) tan 1 2B = (s−c) tan 1 2C, where r2 = (s−a)(s−b)(s−c) s . From these the angles may be found when the sides are given. Given a,b,c , we get the logarithmic relations logr = 1 2 { log(s−a) + log(s−b) + log(s−c)− logs } , and tab log tan1 2A = 10 + logr− log(s−a), tab log tan1 2B = 10 + logr− log(s−b), tab log tan1 2C = 10 + logr− log(s−c), whence the three angles are easily found. There is also a simple veri- fication of the process afforded by taking the sum of the three angles, which ought to be 180◦∗. ∗The formulæ used here were discovered by William Purser of Dublin, in □□□□. Consult Wallace’s Geometrical Theorems.

CHAPTER VIII. OF TRIGONOMETRICAL SURVEYING. In the last Chapter methods have been given of calculating the remaining parts of a triangle when one side and two other parts of the triangle are previously known. In Geodesy, or the application of this part of Trigonometry to sur- veying, a line, called the Base , is measured between two convenient stations; and the angles between lines from these stations to points visible from the stations are measured with appropriate instruments. With these data calculations are made of the distances between new stations, where additional angles may be observed and the operations repeated as often as required. The angles measured by instruments are of three classes. □. Horizontal angles, or angles in the horizontal plane through the station. The horizontal angle between two objects is the angle between the vertical planes through the station and each object: it is also called the angle in azimuth. □. Vertical angles, or angles in a vertical plane through the station. The angle of elevation (or depression) of an object, also called its altitude, is the vertical angle between the horizontal line and the line from the station to the object. □. Angles in planes neither horizontal nor vertical. The angle between two objects is the angle between the lines from the station to each object. This angle is in the plane through the station and the two objects. □□

[Chap. VIII.] OF TRIGONOMETRICAL SURVEYING. □□ In land surveying the angles measured belong to the first or second class, and are measured with a Theodolite or similar instrument. Angles of the third class are usually measured with a Sextant or similar instrument. For details of the measurement of a base, and a description of the instruments principally employed in practical surveying, the reader is referred to Professor Rankine’s Manual of Civil Engineering, Part□. Of Engineering Geodesy, and for the methods employed in a great Trigono- metrical survey to the Ordnance Survey Account of the Principal Tri- angulation, published by the Board of Ordnance, □□□□. Assuming, however, that a base line and angles can be accurately measured by appropriate instruments, it seems within the scope of this elementary treatise to give instances of the determination of heights and distances from such measurements. I. The distance between two stations A and B, visible from each other, being ascertained, and the angles between a visible object C and the other station

being measured at each station, the distances AC and BC can be found by Case I. In this instance the angles measured are in the plane through A, B, C. If the angle of elevation ofC be likewise measured at either station, the height of C above that station can be found by solving a right- angled triangle of which the hypotenuse and one of the acute angles are known. When the horizontal distance of the stations is given and is in a ver- tical plane through C, only vertical angles are required and the work is simplified; and when in addition one station is in the vertical line through C (which is then said to be “accessible”) the measurement of one vertical angle and the solution of one right-angled triangle deter- mine the height of C.

[Chap. VIII.] TRIGONOMETRY. □□ II. From each of two stations A and B, visible from each other, and at a known distance, the angles between the other station and two objects C andD are measured. Then the triangles ABC andABD can be solved (by Case I.), and the distances AC,BC,AD,BD are found. If in addition the angles between C and D are measured at either station, the distance CD can be found by Case II. If the survey extends over so small a fraction of the earth’s surface that the vertical lines over the whole may be considered perpendicular to one horizontal plane, and if the distances measured or calculated are the horizontal distances of the points (or the “projections” of the actual distances on the horizontal plane), which is the case when the survey is conducted for the purpose of making a plan; then the angles to be measured will all be horizontal angles, and the work will be simpler. And if in addition the angle of elevation of any point be observed from a known horizontal distance, the height of the point above the horizontal plane of reference (which may be “the level of the sea”) can be found by the solution of a right-angled triangle. Where the extent of the earth’s surface surveyed is too great for the above supposition, the calculations have to be corrected for the sphericity of the earth (see books on Spherical Trigonometry), and in very great and very accurate surveys also for the earth’s deviation from a perfect sphere. (See the Principal Triangulation of the Ordnance Survey.)

CHAPTER IX. OF PROJECTIONS. The point where the perpendicular from a given point on a given plane or a given line meets the plane or line is called the projection (or more precisely the orthogonal projection) of the point on the plane or line. The part of one line intercepted between the projections on it of the extremities of another line, or of a broken line, is called the projection of the second line, or broken line, on the first. It is convenient to take one direction of the line of projection (say from left to right) as the + direction and the opposite as the− direction; also to consider one end of the projected line as its beginning; and to measure the inclination of the two lines by the angle between the projected line and a line drawn from its beginning parallel to the line of projection and in the + direction. Prop. I. The projection of a line on a given line is equal to the projected line multiplied by the cosine of the angle between the lines. Let AB be the projected line, X′X the line of projection; then in each case, A′B′ being the projection, we have A′B′ =AB cosBAC. If BAC be acute, cos BAC is +, and A′B′ is + and is measured from A′ in the + direction; but if BAC be obtuse, cos BAC is−, and A′B′ is− and is measured from A′ in the− direction. Prop. II. The projection of a broken line is the algebraic sum of the projections of the parts of which it is made up. □□

[Chap. IX.] TRIGONOMETRY. □□ X ′ XA′ A′ A′B′ B′ B′ A A A B B B C C X ′ XA′ B′ C ′ D′ A B C D Let ABCD be a broken line, made up of the parts AB, BC, CD. Let A′B′, B′C′, C′D′ be the projections of these parts on X′X. Thus A′D′ is the projection of the broken line. And A′D′ =A′B′ +B′C′ +C′D′ both in the case where all these projections are + and also where, as in the figure, B′C′ is−.

CHAPTER X. THE SINE AND COSINE OF THE SUM AND DIFFERENCE OF TWO ANGLES. □. To find the sine and cosine of the sum of two angles. A B C D E F Let C be the centre of a circle of which the radius CA =R. Let the circular measure of ACB =θ, and of BCD =φ. Thenθ+φ is the circular measure of ACD. Draw DE, DF perpendicular to CA, CB. Then DE =R sin(θ +φ); CE =R cos(θ +φ); DF =R sinφ; CF =R cosφ. ButDE is the projection of the broken line CFD onDE; therefore DE = projection of CF + projection of FD =CF cos (π 2−θ) +FD cosθ =CF sinθ +FD cosθ, i.e. R sin(θ +φ) =R cosφ sinθ +R sinφ cosθ, or sin(θ +φ) = sinθ cosφ + cosθ sinφ. □□

[Chap. X.] TRIGONOMETRY. □□ Again, CE is the projection of the broken line CFD on CA, □CE = projection of CF + projection of FD =CF cosθ +FD cos (π 2 +θ) =CF cosθ−FD sinθ; i.e. R cos(θ +φ) =R cosφ cosθ−R sinφ sinθ, or cos(θ +φ) = cosθ cosφ− sinθ sinφ. □. To find the sine and cosine of the difference of two angles. A B C D E F Let θ = circular measure of ACB, φ = ... ofBCD . Thenθ−φ = circular measure of ACD. Draw DE, DF perpendicular to CA, CB. Then DE =R sin(θ−φ); CE =R cos(θ−φ); CF =R cosφ; FD =R sinφ. But DE is the projection of CFD on DE = projection of CF + projection of FD =CF cos (π 2−θ) +FD cos(π−θ) =CF sinθ−FD cosθ; i.e. R sin(θ−φ) =R cosφ sinθ−R sinφ cosθ, or sin(θ−φ) = sinθ cosφ− cosθ sinφ.

[Chap. X.] SUM AND DIFFERENCE OF ANGLES □□ Again, CE is the projection of CFD on CA = projection of CF + projection of FD =CF cosθ +FD cos (θ− π 2) =CF cosθ +FD sinθ; i.e. R cos(θ−φ) =R cosφ cosθ +R sinφ sinθ, or cos(θ−φ) = cosφ cosθ + sinφ sinθ. The four expressions for sin(θ±φ) and cos(θ±φ) are true for all values of θ and φ, though the diagrams suppose the angles all acute. They form the fundamental formulæ of Analytical Trigonometry.

APPENDIX. (Note to p. □.) To shew that, when the difference between R and r is small, the limit of each radius is very nearly = 1 3(r + 2R). Let r1,r 2,... be the radii of the circles inscribed in the successive polygons, and R1,R 2,... of those circumscribed about the same. Let also the limit of both be called r∞. Let R−r = 2δ; then r1 = R +r 2 =r +δ =R−δ. Let r2 =r1 +δ1; r3 =r2 +δ2; &c. Then r∞ =r +δ +δ1 +δ2 + &c. ad inf. Also R =r + 2δ; R1 =r1 + 2δ1; &c. and R =r1 +δ; R1 =r2 +δ1; &c. But R2 1 =r1R; or (r1 + 2δ1)2 =r1(r1 +δ); □□

APPENDIX. □□ □ 4r1δ1 + 4δ2 1 =r1δ; □ δ 4− δ2 1 r <δ 1 = δ 4− δ2 1 r1 < δ 4; □ δ 4− 1 r (δ 4)2 <δ 1 < δ 4. Similarly δ1 4− 1 r (δ 42)2 < δ1 4− 1 r (δ1 4)2 <δ 2 < δ1 4, δ2 4− 1 r (δ 43)2 < δ2 4− 1 r (δ2 4)2 <δ 3 < δ2 4, δ3 4− 1 r (δ 44)2 < δ3 4− 1 r (δ3 4)2 <δ 4 < δ3 4, &c. &c. □δ +δ1 +δ2 +... ad inf. <δ + δ 4 + δ1 4 + δ2 4 +... ad inf. and □ 3 4{δ +δ1 +δ2 +... ad inf.}<δ, or δ +δ1 +δ2 +⋯ < 4 3δ. And δ +δ1 +δ2 +⋯ >δ + δ 4 + δ1 4 + &c. ad inf. − 1 r {(δ 4)2 + (δ 42)2 +... ad inf. } ;

TRIGONOMETRY. □□ □ 3 4(δ +δ1 +δ2 +...)>δ − δ2 16r× 16 15 >δ − δ2 15r; □δ +δ1 +δ2 +... ad inf. > 4 3δ− 4 45 δ2 r ; □ 1 3(r + 2R)− (R−r)2 45r <r + 4 3δ− 4 45 δ2

r <r∞ <r + 4 3δ <1 3(r + 2R). So that when, as in the text, R−r <.00003 the error in taking the ultimate radius = 1 3(r + 2R) is < .0000000009 45r which does not affect the tenth decimal place. CAMBRIDGE: PRINTED BY C. J. CLAY, M.A. AT THE UNIVERSITY PRESS.

March, □□□□. A Catalogue of Educational Books with a Short Account of their Character and Aim, Published by MACMILLAN AND CO 16, Bedford Street, Covent Garden, London. CLASSICAL. ÆSCHYLI EUMENIDES. The Greek Text, with English Notes and English Verse, Translation, and an Introduction. By Bernard Drake , M.A., late Fellow of King’s College, Cambridge. 8vo. □s. □d. The Greek text adopted in this Edition is based upon that of Wellauer, which may be said, in general terms, to represent that of the best manuscripts. But in correcting the Text, and in the Notes, advan- tage has been taken of the suggestions of Hermann, Paley, Linwood, and other commentators. In the Translation, the simple character of the æschylean dialogues has generally enabled the author to render them without any material deviation from the construction and idioms of the original Greek. — □ —

EDUCATIONAL BOOKS. ARISTOTLE ON FALLACIES; OR, THE SOPHISTICI ELENCHI. With a Translation and Notes by Edward Poste, M.A., Fellow of Oriel College, Oxford. 8vo. □s. □d. Besides the doctrine of Fallacies, Aristotle offers, either in this trea- tise or in other passages quoted in the commentary, various glances over the world of science and opinion, various suggestions or problems which are still agitated, and a vivid picture of the ancient system of dialectics, which it is hoped may be found both interesting and instructive. Aristotle.—AN INTRODUCTION TO ARISTOTLE’S RHETORIC. With Analysis, Notes, and Appendices. By E. M. Cope , Senior Fellow and Tutor of Trinity College, Cambridge. 8vo. □□s. This work is introductory to an edition of the Greek Text of Aristo- tle’s Rhetoric, which is in course of preparation. Its object is to render that treatise thoroughly intelligible. The author has aimed to illustrate, as preparatory to the detailed explanation of the work, the general bear- ings and relations of the Art of Rhetoric in itself, as well as the special mode of treating it adopted by Aristotle in his peculiar system. The ev- idence upon obscure or doubtful questions connected with the subject is examined; and the relations which Rhetoric bears, in Aristotle’s view, to the kindred art of Logic are fully considered. A connected Analy- sis of the work is given, sometimes in the form of paraphrase; and a few important matters are separately discussed in Appendices. There is added, as a general Appendix, by way of specimen of the antagonistic system of Isocrates and others, a complete analysis of the treatise called <Rhtoriq□ präc > Alèxandron , with a discussion of its authorship and of the probable results of its teaching. Cicero.—THE SECOND PHILIPPIC ORATION. With an Introduction and Notes, translated from the German of Karl Halm . Edited, with Corrections and Additions, by — □ —

CLASSICAL. John E. B. Mayor , M.A., Fellow and Classical Lecturer of St. John’s College, Cambridge. Third Edition, revised. Fcap. 8vo. □s. This volume opens with a List of Books useful to the Student of Cicero, including History, Chronology, Lexicons, and some account of various editions, mostly German, of the works of Cicero. The Intro- duction is based on Halm: where Halm gives a reference to a classic, the

passage has been commonly printed at length; where the reference is to Halm’s notes on other Ciceronian speeches, or to modern books, the additional matter has been incorporated: and the numerous Greek quotations have been rendered into English. The English editor has fur- ther illustrated the work by additions drawn, for the most part, (□) from the ancient authorities; (□) from his own private marginal references, and from collections; (□) from the notes of previous commentators. A copious ‘argument’ is also given. THE ORATIONS OF CICERO AGAINST CATILINA. With Notes translated chiefly from Halm. By A. S. Wilkins , M.A. Fcap. 8vo. □s. □d. This edition is a reprint of the one prepared by Professor Halm for Orelli’s Cicero. The historical introduction of Mr. Wilkins brings together all the details which are known respecting Catiline and his re- lations with the great orator. A list of passages where conjectures have been admitted into the text, and also of all variations from the text of Kayser (□□□□) is added at the end. Finally the English Editor has sub- joined a large number of notes, both original (distinguished by a square bracket) and selected from Curtius, Schleischer, Corssen, and other well-known critics, an analysis of the orations, and an index. DEMOSTHENES ON THE CROWN. The Greek Text with English Notes. By B. Drake , M.A., late Fellow of King’s College, Cambridge. Fourth Edition, to which is prefixed — □ —

EDUCATIONAL BOOKS. ÆSCHINES AGAINST CTESIPHON, with English Notes. Fcap. 8vo. □s. An Introduction discusses the immediate causes of the two orations, and their general character. The Notes contain frequent references to the best authorities. Among the appendices at the end is a chronological table of the life and public career of Æschines and Demosthenes. Hodgson.—MYTHOLOGY FOR LATIN VERSIFICA- TION. A brief Sketch of the Fables of the Ancients, prepared to be rendered into Latin Verse for Schools. By F. Hodgson, B.D., late Provost of Eton. New Edition, revised by F. C. Hodgson , M.A. 18mo. □s. The late Provost of Eton has here supplied a help to the composition of Latin Verse, combined with a brief introduction to Classical Mythol- ogy. In this new edition a few mistakes have been rectified; rules have been added to the Prosody; and a more uniform system has been adopted with regard to the help afforded. Juvenal.—Thirteen Satires of JUVENAL. With a Commentary. By John E. B. Mayor , M.A., Fellow of St. John’s College, Cambridge. Second Edition, enlarged. Part I. Crown 8vo. sewed. □s. □d. The text is accompanied by a copious Commentary. For various notes the author is indebted to Professors Munro and Conington. All the citations have been taken anew from the original authors. Marshall.—A TABLE OF IRREGULAR GREEK VERBS classified according to the arrangement of Curtius’ Greek Grammar. By J. M. Marshall , M.A., Fellow and late Lecturer of Brasenose College, Oxford; one of the Masters in Clifton College. 8vo. cloth. □s. — □ —

CLASSICAL. The system of this table has been borrowed from the excellent Greek Grammar of Dr. Curtius. Mayor (John E. B.) —FIRST GREEK READER. Edited after Karl Halm , with Corrections and large Additions by John E. B. Mayor , M.A. Fellow and Classical Lecturer of St. John’s College, Cambridge. Second and Cheaper Edition. Fcap. 8vo. □s. □d. A selection of short passages, serving to illustrate especially the Greek Accidence. A good deal of syntax is incidentally taught, and Mad- vig and other books are cited, for the use of masters: but no learner is expected to know more of

syntax than is contained in the Notes and Vocabulary. A preface “To the Reader,” not only explains the aim and method of the volume, but also deals with classical instruction gener- ally. The extracts are uniformly in the Attic dialect, and any Hellenis- tic forms occurring in the original classic authors, such as Ælian and Polybius, have been discarded in favour of the corresponding Attic ex- pressions. This book may be used in connexion with Mayor’s “Greek for Beginners.” Mayor (Joseph B.)—GREEK FOR BEGINNERS. By the Rev. J. B. Mayor , M.A., Professor of Classical Literature in King’s College, London. Part I., with Vocabulary, s. d.; Parts II. and III., with Vocabulary and Index, s. d.; complete in one vol., fcap. 8vo. cloth. s. d. The distinctive method of this book consists in building up a boy’s knowledge of Greek upon the foundation of his knowledge of English and Latin, instead of trusting everything to the unassisted memory. The forms and constructions of Greek have been thoroughly compared with those of Latin, and no Greek words have been used in the earlier part of the book except such as have connexions either in English or Latin. Each step leads naturally on to its successor, grammatical forms and rules — —

EDUCATIONAL BOOKS. are at once applied in a series of graduated exercises, accompanied by ample vocabularies. Thus the book serves as Grammar, Exercise book, and Vocabulary. Where possible, the Grammar has been simplified; the ordinary ten declensions are reduced to three, which correspond to the first three in Latin; and the system of stems is adopted. A general Vocabulary, and Index of Greek words, completes the work. Peile (John, M.A.) —AN INTRODUCTION TO GREEK AND LATIN ETYMOLOGY. By John Peile , M.A., Fellow and Assistant Tutor of Christ’s College, Cambridge, formerly Teacher of Sanskrit in the University of Cambridge. 8vo. s. d. These Philological Lectures are the result of Notes made during the author’s reading during the last three or four years. These Notes were put into the shape of lectures, delivered at Christ’s College, during the last May term, as one set in the “Intercollegiate” list. They are now printed with some additions and modifications, but substantially as they were delivered. Plato.—THE REPUBLIC OF PLATO. Translated into English, with an Analysis and Notes, by J. Ll. Davies , M.A., and D. J. Vaughan , M.A. Third Edition, with Vignette Portraits of Plato and Socrates, engraved by Jeens from an Antique Gem. 18mo. s. d. An introductory notice supplies some account of the life of Plato, and the translation is preceded by an elaborate analysis. “The trans- lators have,” in the Judgment of the Saturday Review , “produced a book which any reader, whether acquainted with the original or not, can peruse with pleasure as well as profit.” Plautus (Ramsay). —THE MOSTELLARIA OF PLAUTUS. With Notes Critical and Explanatory, — —

CLASSICAL. Prolegomena, and Excursus. By William Ramsay , M.A., formerly Professor of Humanity in the University of Glasgow. Edited by Professor George G. Ramsay , M.A., of the University of Glasgow. 8vo. s. “The fruits of that exhaustive research and that ripe and well- digested scholarship which its author brought to bear upon everything that he undertook are visible throughout it. It is furnished with a complete apparatus of prolegomena, notes, and excursus; and for the use of veteran scholars it probably leaves nothing to be desired.”— Pall Mall Gazette. Potts (Alex. W., M.A.)—HINTS TOWARDS LATIN PROSE COMPOSITION. By Alex. W. Potts ,

M.A., late Fellow of St. John’s College, Cambridge; Assistant Master in Rugby School; and Head Master of the Fettes College, Edinburgh. Second Edition, enlarged. Extra fcap. 8vo. cloth. s. Those engaged in Classical teaching seem to be unanimously of the opinion that Composition in Latin Prose is not only the most efficient method of acquiring a mastery of the Latin language, but is in itself a valuable means of mental training, and an admirable corrective of some of the worst features in English writing. An attempt is here made to give students, after they have mastered ordinary syntactical rules, some idea of the characteristics of Latin Prose and the means to be employed to reproduce them. Some notion of the treatment of the subject may be gathered from the ‘Contents.’ Chap. I.—Characteristics of Classical Latin, Hints on turning English into Latin; Chap. II. —Arrangement of Words in a Sentence; Chap. III.—Unity in Latin Prose, Subject and Object; Chap. IV.—On the Period in Latin Prose; Chap. V.—On the position of the Relative and Relative Clauses. Roby.—A LATIN GRAMMAR for the Higher Classes in Grammar Schools. By H. J. Roby , M.A. [In the Press. — — —

EDUCATIONAL BOOKS. Sallust.—CAII SALLUSTII CRISPI CATILINA ET JUGURTHA. For Use in Schools. With copious Notes. By C. Merivale, B.D. (In the present Edition the Notes have been carefully revised, and a few remarks and explanations added.) Second Edition. Fcap. 8vo. s. d. The JUGURTHA and the CATILINA may be had separately, price s. d. each. This edition of Sallust, prepared by the distinguished historian of Rome, contains an introduction, concerning the life and works of Sal- lust, lists of the Consuls, and elaborate notes. Tacitus.—THE HISTORY OF TACITUS TRANSLATED INTO ENGLISH. By A. J. Church , M.A., and W. J. Brodribb, M.A. With Notes and a Map. 8vo. s. d. The translators have endeavoured to adhere as closely to the original as was thought consistent with a proper observance of English idiom. At the same time, it has been their aim to reproduce the precise expressions of the author. The campaign of Civilis is elucidated in a note of some length, which is illustrated by a map, containing the names of places and of tribes occurring in the work. There is also a complete account of the Roman army as it was constituted in the time of Tacitus. This work is characterised by the Spectator as “a scholarly and faithful translation.” THE AGRICOLA AND GERMANIA OF TACITUS. A Revised Text, English Notes, and Maps. By Alfred J. Church, M.A., and W. J. Brodribb , M.A. Fcap. 8vo. s. d. “We have endeavoured, with the aid of recent editions, thoroughly to elucidate the text, explaining the various difficulties, critical and gram- matical, which occur to the student. We have consulted throughout, besides the older commentators, the editions of Ritter and Orelli, but — — —

CLASSICAL. we are under special obligations to the labours of the recent German ed- itors, Wex and Kritz.” Two Indexes are appended, () Proper Names, () of Words and Phrases explained. THE AGRICOLA and GERMANIA may be had separately, price s. each. THE AGRICOLA AND GERMANIA. Translated into English by A. J. Church , M.A., and W. J. Brodribb , M.A. With Maps and Notes. Extra fcap. 8vo. s. d. The translators have sought to produce such a version as may satisfy scholars who demand a faithful rendering of the original, and English readers who are offended by the baldness and frigidity which commonly disfigure translations. The treatises are

accompanied by introductions, notes, maps, and a chronological summary. The Athenæum says of this work that it is “a version at once readable and exact, which may be perused with pleasure by all, and consulted with advantage by the classical student.” Theophrastus.—THE CHARACTERS OF THEO- PHRASTUS. An English Translation from a Revised Text. With Introduction and Notes. By R. C. Jebb , M.A., Public Orator in the University of Cambridge. Extra fcap. 8vo. s. d. To the average English reader Theophrastus is little known. At the present time, when there is a general desire to see ancient life more vividly on every side from which it can illustrate our own, it seems possible that the characters of Theophrastus may possess some potent interest. The text has undergone careful revision. An Introduction supplies an account of the origin of the book, and of writers who have imitated it: as Hall, Sir Thomas Overbury, and others. The notes are for the most part selected from ancient sources. — —

EDUCATIONAL BOOKS. Thring.—Works by the Rev. E. THRING, M.A., Head Master of Uppingham School. A LATIN GRADUAL. A First Latin Construing Book for Beginners. By Edward Thring , M.A. New Edition, enlarged, with Coloured Sentence Maps. Fcap. 8vo. s. d. The Head Master of Uppingham has here sought to supply by easy steps a knowledge of grammar, combined with a good Vocabulary. Pas- sages have been selected from the best Latin authors in prose and verse. These passages are gradually built up in their grammatical structure, and finally printed in full. A short practical manual of common mood constructions, with their English equivalents, forms a second part. A MANUAL OF MOOD CONSTRUCTIONS. Fcap. 8vo. s. d. Treats of the ordinary mood constructions, as found in the Latin, Greek, and English languages. A CONSTRUING BOOK. Fcap. 8vo. s. d. Thucydides.—THE SICILIAN EXPEDITION. Being Books VI. and VII. of Thucydides, with Notes. A New Edition, revised and enlarged, with a Map. By the Rev. Percival Frost, M.A., late Fellow of St. John’s College, Cambridge. Fcap. 8vo. s. This edition is mainly a grammatical one. Attention is called to the force of compound verbs, and the exact meaning of the various tenses employed. Virgil.—THE WORKS OF VIRGIL RENDERED INTO ENGLISH PROSE , with Introductions, Running Analysis, and an Index, by James Lonsdale , M.A. and Samuel Lee, M.A. Globe 8vo. s. d.; gilt edges, s. d. — —

CLASSICAL. The preface of this new volume informs us that “the original has been faithfully rendered, and paraphrase altogether avoided. At the same time, the translators have endeavoured to adapt the book to the use of the English reader. Some amount of rhythm in the structure of the sentence has been generally maintained; and, when in the Latin the sound of the words is an echo to the sense (as so frequently happens in Virgil), an attempt has been made to produce the same result in English.” The general introduction gives us whatever is known of the poet’s life, an estimate of his genius, an account of the principal editions and translations of his works, and a brief view of the influence he has had on modern poets; special introductory essays are prefixed to the “Eclogues,” “Georgics,” and “Æneid.” The text is divided into sections, each of which is headed by a concise analysis of the subject; the index contains references to all the characters and events of any importance. Wright.—Works by J. Wright , M.A., late Head Master of Sutton Coldfield School.

HELLENICA; OR, A HISTORY OF GREECE IN GREEK, as related by Diodorus and Thucydides; being a First Greek Reading Book, with explanatory Notes, Critical and Historical. Third Edition, with a Vocabulary. 12mo. s. d. In the last twenty chapters of this volume, Thucydides sketches the rise and progress of the Athenian Empire in so clear a style and in such simple language, that the editor has doubts whether any easier or more instructive passages can be selected for the use of the pupil who is commencing Greek. This book includes a chronological table of the events recorded. A HELP TO LATIN GRAMMAR ; or, The Form and Use of Words in Latin, with Progressive Exercises. Crown 8vo. s. d. — —

EDUCATIONAL BOOKS. This book is not intended as a rival to any of the excellent Grammars now in use; but as a help to enable the beginner to understand them. THE SEVEN KINGS OF ROME. An Easy Narrative, abridged from the First Book of Livy by the omission of Difficult Passages; being a First Latin Reading Book, with Grammatical Notes. With Vocabulary and Exercises. Fourth Edition. Fcap. 8vo. s. This work is intended to supply the pupil with an easy construing book, which may at the same time be made the vehicle for instructing him in the rules of grammar and principles of composition. The notes profess to teach what is commonly taught in grammars. It is conceived that the pupil will learn the rules of construction of the language much more easily from separate examples, which are pointed out to him in the course of his reading, and which he may himself set down in his note- book after some scheme of his own, than from a heap of quotations amassed for him by others. Or, separately, SEVEN KINGS OF ROME. s. VOCABULARY AND EXERCISES TO “THE SEVEN KINGS.” s. d. CLASSIC VERSIONS OF ENGLISH BOOKS, AND LATIN HYMNS. The following works are, as the heading indicates, classic renderings of English books. For scholars, and particularly for writers of Latin Verse, the series has a special value. The Hymni Ecclesiæ are here inserted, as partly falling under the same class. — —

CLASSICAL. Church (A. J., A.M.) —HORÆ TENNYSONIANÆ , sive Eclogae e Tennysono. Latine redditæ. Cura A. J. Church, A.M. Extra fcap. 8vo. s. Latin versions of Selections from Tennyson. Among the authors are the Editor, the late Professor Conington, Professor Seeley, Dr. Hessey, Mr. Kebbel, and other gentlemen. Latham.—SERTUM SHAKSPERIANUM , Subnexis aliquot aliunde excerptis floribus. Latine reddidit Rev. H. Latham, M.A. Extra fcap. 8vo. s. Besides versions of Shakespeare this volume contains, among other pieces, Gray’s “Elegy,” Campbell’s “Hohenlinden,” Wolfe’s “Burial of Sir John Moore,” and selections from Cowper and George Herbert. Lyttelton.—THE COMUS OF MILTON , rendered into Greek Verse. By Lord Lyttelton. Extra fcap. 8vo. s. THE SAMSON AGONISTES OF MILTON , rendered into Greek Verse. By Lord Lyttelton . Extra fcap. 8vo. s. d. Merivale.—KEATS’ HYPERION, rendered into Latin Verse. By C. Merivale , B.D. Second Edit. Extra fcap. 8vo. s. d. Hymni Ecclesiæ.—Edited by Rev. Dr. Newman. Extra fcap. 8vo. s. d. Hymns of the Mediæval Church. The first Part contains selections from the Parisian Breviary; the second from those of Rome, Salisbury, and York. — —

EDUCATIONAL BOOKS. Trench (Archbishop).—SACRED LATIN POETRY , chiefly Lyrical, selected and arranged for Use; with Notes and Introduction. Fcap. 8vo. s. In this work the editor has selected hymns of a catholic religious sentiment that are common to Christendom, while rejecting those of a distinctively Romish character. — —

MATHEMATICS. MATHEMATICS. Airy.—Works by G. B. AIRY, Astronomer Royal:— ELEMENTARY TREATISE ON PARTIAL DIFFERENTIAL EQUATIONS. Designed for the Use of Students in the Universities. With Diagrams. Crown 8vo. cloth. s. d. It is hoped that the methods of solution here explained, and the instances exhibited, will be found sufficient for application to nearly all the important problems of Physical Science, which require for their complete investigation the aid of Partial Differential Equations. ON THE ALGEBRAICAL AND NUMERICAL THEORY OF ERRORS OF OBSERVATIONS AND THE COMBINATION OF OBSERVATIONS. Crown 8vo. cloth. s. d. In order to spare astronomers and observers in natural philosophy the confusion and loss of time which are produced by referring to the ordinary treatises embracing both branches of probabilities (the first relating to chances which can be altered only by the changes of entire units or integral multiples of units in the fundamental conditions of the problem; the other concerning those chances which have respect to insensible gradations in the value of the element measured) the present tract has been drawn up. It relates only to errors of observation, and to the rules, derivable from the consideration of these errors, for the combination of the results of observations. UNDULATORY THEORY OF OPTICS. Designed for the Use of Students in the University. New Edition. Crown 8vo. cloth. s. d. The undulatory theory of optics is presented to the reader as having the same claims to his attention as the theory of gravitation: namely, — —

EDUCATIONAL BOOKS. that it is certainly true, and that, by mathematical operations of general elegance, it leads to results of great interest. This theory explains with accuracy a vast variety of phenomena of the most complicated kind. The plan of this tract has been to include those phenomena only which admit of calculation, and the investigations are applied only to phenomena which actually have been observed. ON SOUND AND ATMOSPHERIC VIBRATIONS. With the Mathematical Elements of Music. Designed for the Use of Students of the University. Crown 8vo. s. This volume consists of sections, which again are divided into numbered articles, on the following topics: General recognition of the air as the medium which conveys sound; Properties of the air on which the formation and transmission of sound depend; Theory of undulations as applied to sound, &c.; Investigation of the motion of a wave of air through the atmosphere; Transmission of waves of soniferous vibrations through different gases, solids, and fluids; Experiments on the velocity of sound, &c.; On musical sounds, and the manner of producing them; On the elements of musical harmony and melody, and of simple musical composition; On instrumental music; On the human organs of speech and hearing. A TREATISE ON MAGNETISM. Designed for the use of Students in the University. Crown 8vo. s. d. Airy (Osmund.)—A TREATISE ON GEOMETRICAL OPTICS. Adapted for the use of the Higher Classes in Schools. By Osmund Airy , B.A., one of the Mathematical Masters in Wellington College. Extra fcap. 8vo. s. d. “This is, I imagine, the first time that any

attempt has been made to adapt the subject of Geometrical Optics, to the reading of the higher classes in our good schools. That this should be so is the more a mat- ter for remark, since the subject would appear to be peculiarly fitted for — □□ —

MATHEMATICS. such an adaptation. . . . I have endeavoured, as much as possible, to avoid the example of those popular lecturers who explain difficulties by ignoring them. But as the nature of my design necessitated brevity, I have omitted entirely one or two portions of the subject which I consid- ered unnecessary to a clear understanding of the rest, and which appear to me better learnt at a more advanced stage."— Author's Preface. Bayma.—THE ELEMENTS OF MOLECULAR MECHAN- ICS. By Joseph Bayma , S.J., Professor of Philosophy, Stonyhurst College. Demy 8vo. cloth. □□s. □d. Of the twelve Books into which the present treatise is divided, the first and second give the demonstration of the principles which bear di- rectly on the constitution and the properties of matter. The next three books contain a series of theorems and of problems on the laws of mo- tion of elementary substances. In the sixth and seventh, the mechanical constitution of molecules is investigated and determined: and by it the general properties of bodies are explained. The eighth book treats of lu- miniferous æther. The ninth explains some special properties of bodies. The tenth and eleventh contain a radical and lengthy investigation of chemical principles and relations, which may lead to practical results of high importance. The twelfth and last book treats of molecular masses, distances, and powers. Beasley.—AN ELEMENTARY TREATISE ON PLANE TRIGONOMETRY. With Examples. By R. D. Beasley , M.A., Head Master of Grantham Grammar School. Second Edition, revised and enlarged. Crown 8vo. cloth. □s. □d. This treatise is specially intended for use in schools. The choice of matter has been chiefly guided by the requirements of the three days' examination at Cambridge. About four hundred examples have been added to this edition, mainly collected from the Examination Papers of the last ten years. — □□ —

EDUCATIONAL BOOKS. Boole.—Works by G. BOOLE, D.C.L., F.R.S., Professor of Mathematics in the Queen's University, Ireland. A TREATISE ON DIFFERENTIAL EQUATIONS. New and Revised Edition. Edited by I. Todhunter . Crown 8vo. cloth. □□s. Professor Boole has endeavoured in this treatise to convey as com- plete an account of the present state of knowledge on the subject of Differential Equations, as was consistent with the idea of a work in- tended, primarily, for elementary instruction. The earlier sections of each chapter contain that kind of matter which has usually been thought suitable for the beginner, while the latter ones are devoted either to an account of recent discovery, or the discussion of such deeper questions of principle as are likely to present themselves to the reflective student in connexion with the methods and processes of his previous course. A TREATISE ON DIFFERENTIAL EQUATIONS. Supple- mentary Volume. Edited by I. Todhunter . Crown 8vo. cloth. □s. □d. This volume contains all that Professor Boole wrote for the purpose of enlarging his treatise on Differential Equations. THE CALCULUS OF FINITE DIFFERENCES. Crown 8vo. cloth. □□s. □d. In this exposition of the Calculus of Finite Differences, particular attention has been paid to the connexion of its methods with those of the Differential Calculus—a connexion which in some instances involves far more than a merely formal analogy. The work is in some measure

designed as a sequel to Professor Boole’s Treatise on Differential Equa- tions. — □□ —

MATHEMATICS. CAMBRIDGE SENATE-HOUSE PROBLEMS AND RIDERS, WITH SOLUTIONS :— □□□□-□□□□.—PROBLEMS. By Ferrers and Jackson. 8vo. cloth. □□s. □d. □□□□-□□□□.—RIDERS. By Jameson. 8vo. cloth. □s. □d. □□□□.—PROBLEMS AND RIDERS. By Walton and Mackenzie. 8vo. cloth. □□s. □d. □□□□.—PROBLEMS AND RIDERS. By Campion and Walton. 8vo. cloth. □s. □d. □□□□.—PROBLEMS AND RIDERS. By Watson and Routh. Crown 8vo. cloth. □s. □d. □□□□.—PROBLEMS AND RIDERS. By Walton and Wilkinson. 8vo. cloth. □□s. □d. These volumes will be found of great value to Teachers and Students, as indicating the style and range of mathematical study in the University of Cambridge. CAMBRIDGE COURSE OF ELEMENTARY NATURAL PHILOSOPHY, for the Degree of B.A. Originally compiled by J. C. Snowball , M.A., late Fellow of St. John’s College. Fifth Edition, revised and enlarged, and adapted for the Middle-Class Examinations by Thomas Lund , B.D., Late Fellow and Lecturer of St. John’s College, Editor of Wood’s Algebra, &c. Crown 8vo. cloth. □s. This work will be found adapted to the wants, not only of Univer- sity Students, but also of many others who require a short course of Mechanics and Hydrostatics, and especially of the candidates at our Middle Class Examinations. At the end of each chapter a series of easy questions is added for the exercise of the student. — □□ —

EDUCATIONAL BOOKS. CAMBRIDGE AND DUBLIN MATHEMATICAL JOURNAL. The Complete Work, in Nine Vols. 8vo. cloth, □l. □s. Only a few copies remain on hand. Among Contributors to this work will be found Sir W. Thomson, Stokes, Adams, Boole, Sir W. R. Hamil- ton, De Morgan, Cayley, Sylvester, Jellett, and other distinguished mathematicians. Candler.—HELP TO ARITHMETIC. Designed for the use of Schools. By H. Candler , M.A. Mathematical Master of Uppingham School. Extra fcap. 8vo. □s. □d. This work is intended as a companion to any text book that may be in use. Cheyne.—AN ELEMENTARY TREATISE ON THE PLANETARY THEORY. With a Collection of Problems. By C. H. H. Cheyne , M.A., F.R.A.S. Second Edition. Crown 8vo. cloth. □s. □d. In this volume, an attempt has been made to produce a treatise on the Planetary theory, which, being elementary in character, should be so far complete, as to contain all that is usually required by students in the University of Cambridge. This Edition has been carefully revised. The stability of the Planetary System has been more fully treated, and an elegant geometrical explanation of the formulæ for the secular vari- ation of the node and inclination, due to Mr. H. M. Taylor, has been introduced. THE EARTH’S MOTION OF ROTATION. By C. H. H. Cheyne, M.A., F.R.A.S. Crown 8vo. □s. □d. The first part of this work consists of an application of the method of the variation of elements to the general problem of rotation. In the second part the general rotation formulæ are applied to the particular case of the earth. — □□ —

MATHEMATICS. Childe.—THE SINGULAR PROPERTIES OF THE ELLIPSOID AND ASSOCIATED SURFACES OF THE Nth DEGREE. By the Rev. G. F. Childe , M.A., Author of “Ray Surfaces,” “Related Caustics,” &c. 8vo. □□s. □d. The object of this volume is to develop peculiarities in the Ellip- soid; and, further, to establish analogous properties in the unlimited congeneric series of which this remarkable

surface is a constituent. Christie.—A COLLECTION OF ELEMENTARY TEST-QUESTIONS IN PURE AND MIXED MATHEMATICS; with Answers and Appendices on Synthetic Division, and on the Solution of Numerical Equations by Horner's Method. By James R. Christie , F.R.S., late First Mathematical Master at the Royal Military Academy, Woolwich. Crown 8vo. cloth. s. d. The series of Mathematical exercises here offered to the public is collected from those which the author has, from time to time, proposed for solution by his pupils during a long career at the Royal Military Academy. A student who finds that he is able to solve the larger portion of these exercises, may consider that he is thoroughly well grounded in the elementary principles of pure and mixed Mathematics. Dalton.—ARITHMETICAL EXAMPLES. Progressively ar- ranged, with Exercises and Examination Papers. By the Rev. T. Dalton , M.A., Assistant Master of Eton College, 18mo. cloth. s. d. Answers to the Examples are appended. Day.—PROPERTIES OF CONIC SECTIONS PROVED GEOMETRICALLY. PART I., THE ELLIPSE , with Problems. By the Rev. H. G. Day , M.A., Head Master of Sedburgh Grammar School. Crown 8vo. s. d. — —

EDUCATIONAL BOOKS. The object of this book is the introduction of a treatment of Conic Sections which should be simple and natural, and lead by an easy tran- sition to the analytical methods, without departing from the strict ge- ometry of Euclid. Dodgson.—AN ELEMENTARY TREATISE ON DETER- MINANTS, with their Application to Simultaneous Linear Equations and Algebraical Geometry. By Charles L. Dodgson, M.A., Student and Mathematical Lecturer of Christ Church, Oxford. Small 4to. cloth. s. d. The object of the author is to present the subject as a continuous chain of argument, separated from all accessories of explanation or il-lustration. All such explanation and illustration as seemed necessary for a beginner are introduced either in the form of foot-notes, or, where that would have occupied too much room, of Appendices. Drew.—GEOMETRICAL TREATISE ON CONIC SEC-TIONS. By W. H. Drew , M.A., St. John's College, Cambridge. Fourth Edition. Crown 8vo. cloth. s. d. In this work the subject of Conic Sections has been placed before the student in such a form that, it is hoped, after mastering the elements of Euclid, he may find it an easy and interesting continuation of his geometrical studies. With a view, also, of rendering the work a complete manual of what is required at the Universities, there have either been embodied into the text or inserted among the examples, every book-work question, problem, and rider, which has been proposed in the Cambridge examinations up to the present time. SOLUTIONS TO THE PROBLEMS IN DREW'S CONIC SECTIONS. Crown 8vo. cloth. s. d. — —

MATHEMATICS. Edgar (J. H.)—NOTE-BOOK ON PRACTICAL SOLID GEOMETRY. Containing Problems with help for Solutions. By J. H. Edgar , M.A. Lecturer on Mechanical Drawing at the Royal School of Mines. 4to. s. In teaching a large class, if the method of lecturing and demonstrat- ing from the black board only is pursued, the more intelligent students have generally to be kept back, from the necessity of frequent repetition, for the sake of the less promising; if the plan of setting problems to each pupil is adopted, the teacher finds a difficulty in giving to each sufficient attention. A judicious combination of both methods is doubt- less the best; and it is hoped that this result may be arrived at in some degree by the use of this book, which is simply a collection of examples, with helps for solution, arranged in progressive

sections. Ferrers.—AN ELEMENTARY TREATISE ON TRILIN- EAR CO-ORDINATES, the Method of Reciprocal Polars, and the Theory of Projectors. By the Rev. N. M. Ferrers , M.A., Fellow and Tutor of Gonville and Caius College, Cambridge. Second Edition. Crown 8vo. s. d. The object of the author in writing on this subject has mainly been to place it on a basis altogether independent of the ordinary Cartesian sys- tem, instead of regarding it as only a special form of Abridged Notation. A short chapter on Determinants has been introduced. Frost.—THE FIRST THREE SECTIONS OF NEWTON'S PRINCIPIA. With Notes and Illustrations. Also a collection of Problems, principally intended as Examples of Newton's Methods. By Percival Frost , M.A., late Fellow of St. John's College, Mathematical Lecturer of King's College, Cambridge. Second Edition. 8vo. cloth. s. d. The author's principal intention is to explain difficulties which may be encountered by the student on first reading the Principia, and to il- lustrate the advantages of a careful study of the methods employed by — —

EDUCATIONAL BOOKS. Newton, by showing the extent to which they may be applied in the so- lution of problems; he has also endeavoured to give assistance to the student who is engaged in the study of the higher branches of mathe- matics, by representing in a geometrical form several of the processes employed in the Differential and Integral Calculus, and in the analytical investigations of Dynamics. Frost and Wolstenholme. —A TREATISE ON SOLID GEOMETRY. By Percival Frost , M.A., and the Rev. J. Wolstenholme , M.A., Fellow and Assistant Tutor of Christ's College. 8vo. cloth. s. The authors have endeavoured to present before students as compre- hensive a view of the subject as possible. Intending to make the subject accessible, at least in the earlier portion, to all classes of students, they have endeavoured to explain completely all the processes which are most useful in dealing with ordinary theorems and problems, thus directing the student to the selection of methods which are best adapted to the exigencies of each problem. In the more difficult portions of the sub- ject, they have considered themselves to be addressing a higher class of students; and they have there tried to lay a good foundation on which to build, if any reader should wish to pursue the science beyond the limits to which the work extends. Godfray.—A TREATISE ON ASTRONOMY , for the Use of Colleges and Schools. By Hugh Godfray , M.A., Mathematical Lecturer at Pembroke College, Cambridge. 8vo. cloth. s. d. This book embraces all those branches of Astronomy which have, from time to time, been recommended by the Cambridge Board of Math- ematical Studies: but by far the larger and easier portion, adapted to the first three days of the Examination for Honours, may be read by the more advanced pupils in many of our schools. The author's aim has been to convey clear and distinct ideas of the celestial phenomena. — —

MATHEMATICS. AN ELEMENTARY TREATISE ON THE LUNAR THEORY, with a Brief Sketch of the Problem up to the time of Newton. By Hugh Godfray, M.A. Second Edition, revised. Crown 8vo. cloth. s. d. These pages will, it is hoped, form an introduction to more recondite works. Difficulties have been discussed at considerable length. The se- lection of the method followed with regard to analytical solutions, which is the same as that of Airy, Herschel, &c. was made on account of its simplicity; it is, moreover, the method which has obtained in the University of

Cambridge. Hemming.—AN ELEMENTARY TREATISE ON THE DIFFERENTIAL AND INTEGRAL CALCULUS , for the Use of Colleges and Schools. By G. W. Hemming , M.A., Fellow of St. John’s College, Cambridge. Second Edition, with Corrections and Additions. 8vo. cloth. s. Jones and Cheyne. —ALGEBRAICAL EXERCISES. Progressively arranged. By the Rev. C. A. Jones , M.A., and C. H. Cheyne , M.A., F.R.A.S., Mathematical Masters of Westminster School. New Edition. 18mo. cloth. s. d. This little book is intended to meet a difficulty which is probably felt more or less by all engaged in teaching Algebra to beginners. It is, that while new ideas are being acquired, old ones are forgotten. In the belief that constant practice is the only remedy for this, the present series of miscellaneous exercises has been prepared. Their peculiarity consists in this, that though miscellaneous they are yet progressive, and may be used by the pupil almost from the commencement of his studies. They are not intended to supersede the systematically arranged examples to be found in ordinary treatises on Algebra, but rather to supplement them. The book being intended chiefly for Schools and Junior Students, the higher parts of Algebra have not been included. — — —

EDUCATIONAL BOOKS. Kitchener.—A GEOMETRICAL NOTE-BOOK, containing Easy Problems in Geometrical Drawing preparatory to the Study of Geometry. For the Use of Schools. By F. E. Kitchener, M.A., Mathematical Master at Rugby. 4to. s. It is the object of this book to make some way in overcoming the difficulties of Geometrical conception, before the mind is called to the attack of Geometrical theorems. A few simple methods of construction are given; and space is left on each page, in order that the learner may draw in the figures. Morgan.—A COLLECTION OF PROBLEMS AND EXAMPLES IN MATHEMATICS. With Answers. By H. A. Morgan , M.A., Sadlerian and Mathematical Lecturer of Jesus College, Cambridge. Crown 8vo. cloth. s. d. This book contains a number of problems, chiefly elementary, in the Mathematical subjects usually read at Cambridge. They have been selected from the papers set during late years at Jesus College. Very few of them are to be met with in other collections, and by far the larger number are due to some of the most distinguished Mathematicians in the University. Parkinson.—Works by S. Parkinson , D.D., F.R.S., Fellow and Tutor of St. John’s College, Cambridge. AN ELEMENTARY TREATISE ON MECHANICS. For the Use of the Junior Classes at the University and the Higher Classes in Schools. With a Collection of Examples. Fourth edition, revised. Crown 8vo. cloth. s. d. In preparing a fourth edition of this work the author has kept the same object in view as he had in the former editions—namely, to in- clude in it such portions of Theoretical Mechanics as can be conveniently investigated without the use of the Differential Calculus, and so render — — —

MATHEMATICS. it suitable as a manual for the junior classes in the University and the higher classes in Schools. With one or two short exceptions, the student is not presumed to require a knowledge of any branches of Mathematics beyond the elements of Algebra, Geometry, and Trigonometry. Several additional propositions have been incorporated in the work for the pur- pose of rendering it more complete; and the collection of Examples and Problems has been largely increased. A TREATISE ON OPTICS. Third Edition, revised and enlarged. Crown 8vo. cloth. s. d. A collection of examples and problems has been appended to this work, which are

sufficiently numerous and varied in character to afford useful exercise for the student. For the greater part of them, recourse has been had to the Examination Papers set in the University and the several Colleges during the last twenty years. Phear.—ELEMENTARY HYDROSTATICS.With Numerous Examples. By J. B. Phear , M.A., Fellow and late Assistant Tutor of Clare College, Cambridge. Fourth Edition. Crown 8vo. cloth. s. d. This edition has been carefully revised throughout, and many new illustrations and examples added, which it is hoped will increase its usefulness to students at the Universities and in Schools. In accor- dance with suggestions from many engaged in tuition, answers to all the Examples have been given at the end of the book. Pratt.—A TREATISE ON ATTRACTIONS, LAPLACE'S FUNCTIONS, AND THE FIGURE OF THE EARTH. By John H. Pratt , M.A., Archdeacon of Calcutta, Author of "The Mathematical Principles of Mechanical Philosophy." Third Edition. Crown 8vo. cloth. s. d. — —

EDUCATIONAL BOOKS. The author's chief design in this treatise is to give an answer to the question, "How the Earth acquired its present form from being originally in a fluid state?" This Edition is a complete revision of the former ones. Puckle.—AN ELEMENTARY TREATISE ON CONIC SECTIONS AND ALGEBRAIC GEOMETRY. With Numerous Examples and Hints for their Solution; especially designed for the Use of Beginners. By G. H. Puckle , M.A., Head Master of Windermere College. New Edition, revised and enlarged. Crown 8vo. cloth. s. d. This work is recommended by the Syndicate of the Cambridge Local Examinations, and is the text-book in Harvard University, U.S. Rawlinson.—ELEMENTARY STATICS , by the Rev. GEORGE RAWLINSON , M.A. Edited by the Rev. Edward Sturges, M.A., of Emmanuel College, Cambridge, and late Professor of the Applied Sciences, Elphinstone College, Bombay. Crown 8vo. cloth. s. d. Published under the authority of Her Majesty's Secretary of State for India, for use in the Government Schools and Colleges in India. Reynolds.—MODERN METHODS IN ELEMENTARY GEOMETRY. By E. M. Reynolds , M.A., Mathematical Master in Clifton College. Crown 8vo. s. d. Some change, it is evident, in our English ways of teaching can now no longer be postponed, and this little book, mainly derived from French and German sources, has been written in the hope of facilitat- ing that change. It has been constructed on one plan throughout, that of always giving in the simplest possible form the direct proof from the nature of the case. The axioms necessary to this simplicity have been assumed without hesitation, and no scruple has been felt as to the in- crease of their number, or the acceptance of as many elementary notions as common experience places past all doubt. — —

MATHEMATICS. The book differs most from established teaching in its constructions, and in its early application of Arithmetic to Geometry. Routh.—AN ELEMENTARY TREATISE ON THE DYNAMICS OF THE SYSTEM OF RIGID BODIES. With Numerous Examples. By Edward John Routh , M.A., late Fellow and Assistant Tutor of St. Peter's College, Cambridge; Examiner in the University of London. Second Edition, enlarged. Crown 8vo. cloth. s. In this edition the author has made several additions to each chapter. He has tried, even at the risk of some little repetition, to make each chapter, as far as possible, complete in itself, so that all that relates to any one part of the subject may be found in the same place. This

arrangement will enable every student to select his own order in which to read the subject. The Examples which will be found at the end of each chapter have been chiefly selected from the Examination Papers which have been set in the University and the Colleges in the last few years. Smith (Barnard).—Works by BARNARD SMITH, M.A., Rector of Glaston, Rutlandshire, late Fellow and Senior Bursar of St. Peter’s College, Cambridge. ARITHMETIC AND ALGEBRA , in their Principles and Application; with numerous systematically arranged Examples taken from the Cambridge Examination Papers, with especial reference to the Ordinary Examination for the B.A. Degree. Tenth Edition. Crown 8vo. cloth. □□s. □d. This manual is now extensively used in Schools and Colleges, both in England and in the Colonies. It has also been found of great service for students preparing for the Middle Class and Civil and Military Service Examinations, from the care that has been taken to elucidate the prin- ciples of all the rules. The present edition has been carefully revised. — □□ —

EDUCATIONAL BOOKS. “To all those whose minds are sufficiently developed to comprehend the simplest mathematical reasoning, and who have not yet thoroughly mas- tered the principles of Arithmetic and Algebra, it is calculated to be of great advantage.”—Athenæum. Of this work, also, one of the highest possible authorities, the late Dean Peacock, writes: “Mr. Smith’s work is a most useful publication. The rules are stated with great clearness. The examples are well se- lected, and worked out with just sufficient detail, without being encum- bered by too minute explanations; and there prevails throughout it that just proportion of theory and practice, which is the crowning excellence of an elementary work.” ARITHMETIC FOR SCHOOLS. New Edition. Crown 8vo. cloth. □s. □d. Adapted from the author’s work on “Arithmetic and Algebra,” by the omission of the algebraic portion, and by the introduction of new exercises. The reason of each arithmetical process is fully exhibited. The system of Decimal Coinage is explained; and answers to the exercises are appended at the end. This Arithmetic is characterised as “admirably adapted for instruction, combining just sufficient theory with a large and well-selected collection of exercises for practice.”— Journal of Education. COMPANION TO ARITHMETIC FOR SCHOOLS. [Preparing. A KEY TO THE ARITHMETIC FOR SCHOOLS. Seventh Edition. Crown 8vo. cloth. □s. □d. EXERCISES IN ARITHMETIC. With Answers. Crown 8vo. limp cloth. □s. □d. Or sold separately, Part I. □s.; Part II. □s.; Answers, □d. — □□ —

MATHEMATICS. These Exercises have been published in order to give the pupil exam- ples in every rule of Arithmetic. The greater number have been carefully compiled from the latest University and School Examination Papers. SCHOOL CLASS-BOOK OF ARITHMETIC. 18mo. cloth. □s. Or sold separately, Parts I. and II. □□d. each; Part III. □s. This manual, published at the request of many schoolmasters, and chiefly intended for National and Elementary Schools, has been prepared on the same plan as that adopted in the author’s School Arithmetic, which is in extensive circulation in England and abroad. The Metrical Tables have been introduced, from the conviction on the part of the author, that the knowledge of such tables, and the mode of applying them, will be of great use to the rising generation. KEYS TO SCHOOL CLASS-BOOK OF ARITHMETIC. Complete in one volume, 18mo. cloth, □s. □d.; or Parts I. II. and III. □s. □d. each. SHILLING BOOK OF ARITHMETIC FOR

NATIONAL AND ELEMENTARY SCHOOLS. 18mo. cloth. Or separately, Part I. d.; Part II. d.; Part III. d. Answers, d. THE SAME , with Answers complete. 18mo. cloth. s. d. This Shilling Book of Arithmetic has been prepared for the use of National and other schools at the urgent request of numerous masters of schools both at home and abroad. The Explanations of the Rules, and the Examples will, it is hoped, be found suited to the most elementary classes. KEY TO SHILLING BOOK OF ARITHMETIC. 18mo. cloth. s. d. — —

EDUCATIONAL BOOKS. EXAMINATION PAPERS IN ARITHMETIC. 18mo. cloth. s. d. The same, with Answers, 18mo. s. d. The object of these Examination Papers is to test students both in the theory and practice of Arithmetic. It is hoped that the method adopted will lead students to deduce results from general principles rather than to apply stated rules. The author believes that the practice of giving examples under particular rules makes the working of Arith- metic quite mechanical, and tends to throw all but very clever boys off their balance when a general paper on the subject is put before them. KEY TO EXAMINATION PAPERS IN ARITHMETIC. 18mo. cloth. s. d. Smith (J. Brook).—ARITHMETIC IN THEORY AND PRACTICE, FOR ADVANCED PUPILS. By J. Brook Smith, M.A. Part I. Crown 8vo. s. d. The following pages form the first part of a Treatise on Arithmetic, in which the Author has endeavoured from very simple principles to explain, in a full and satisfactory manner, all the more important pro- cesses in that subject. The proofs have in all cases been given in a form entirely arithmetical, and at the end of every chapter several ex- amples have been worked out at length, and the best practical method of operation carefully pointed out. Snowball.—THE ELEMENTS OF PLANE AND SPHER- ICAL TRIGONOMETRY; with the Construction and Use of Tables of Logarithms. By J. C. Snowball , M.A. Tenth Edition. Crown 8vo. cloth. s. d. In preparing the present edition for the press, the text has been sub- jected to a careful revision; the proofs of some of the more important propositions have been rendered more strict and general; and a consid- erable addition of more than two hundred examples, taken principally — —

MATHEMATICS. from the questions set of late years in the public examinations of the University and of individual Colleges, has been made to the collection of Examples and Problems for practice. Tait and Steele.—A TREATISE ON DYNAMICS OF A PARTICLE. With numerous Examples. By Professor Tait and Mr. Steele. New Edition Enlarged. Crown 8vo. cloth. s. d. In this treatise will be found all the ordinary propositions, connected with the Dynamics of Particles, which can be conveniently deduced with- out the use of D’Alembert’s Principle. Throughout the book will be found a number of illustrative examples introduced in the text, and for the most part completely worked out; others with occasional solutions or hints to assist the student are appended to each chapter. For by far the greater portion of these, the Cambridge Senate-House and College Examination Papers have been applied to. Taylor.—GEOMETRICAL CONICS ; including Anharmonic Ratio and Projection, with numerous Examples. By C. Taylor, B.A., Scholar of St. John’s College, Cambridge. Crown 8vo. cloth. s. d. This work contains elementary proofs of the principal properties of Conic Sections, together with chapters on Projection and Anharmonic Ratio. Tebay.—ELEMENTARY MENSURATION FOR SCHOOLS. With

numerous Examples. By Septimus Tebay, B.A., Head Master of Queen Elizabeth's Grammar School, Rivington. Extra fcap. 8vo. s. d. The object of the present work is to enable boys to acquire a moderate knowledge of Mensuration in a reasonable time. All difficult and useless matter has been avoided. The examples for the most part are easy, and the rules are concise. — —

EDUCATIONAL BOOKS. Todhunter.—Works by I. Todhunter , M.A., F.R.S., of St. John's College, Cambridge. THE ELEMENTS OF EUCLID. For the Use of Colleges and Schools. New Edition. 18mo. cloth. s. d. As the elements of Euclid are usually placed in the hands of young students, it is important to exhibit the work in such a form as will assist them in overcoming the difficulties which they experience on their first introduction to processes of continuous argument. No method appears to be so useful as that of breaking up the demonstrations into their con- stituent parts; a plan strongly recommended by Professor De Morgan. In the present Edition each distinct assertion in the argument begins a new line: and at the ends of the lines are placed the necessary refer- ences to the preceding principles on which the assertions depend. The longer propositions are distributed into subordinate parts, which are distinguished by breaks at the beginning of the lines. Notes, appendix, and a collection of exercises are added. MENSURATION FOR BEGINNERS. With Numerous Examples. 18mo. cloth. s. d. The subjects included in the present work are those which have usu- ally found a place in Elementary Treatises on Mensuration. The mode of treatment has been determined by the fact that the work is intended for the use of beginners. Accordingly it is divided into short independent chapters, which are followed by appropriate examples. A knowledge of the elements of Arithmetic is all that is assumed; and in connexion with most of the Rules of Mensuration it has been found practicable to give such explanations and illustrations as will supply the place of formal mathematical demonstrations, which would have been unsuitable to the character of the work. ALGEBRA FOR BEGINNERS. With numerous Examples. New Edition. 18mo. cloth. s. d. — —

MATHEMATICS. Great pains have been taken to render this work intelligible to young students, by the use of simple language and by copious explanations. In determining the subjects to be included and the space to be assigned to each, the Author has been guided by the papers given at the vari- ous examinations in elementary Algebra which are now carried on in this country. The book may be said to consist of three parts. The first part contains the elementary operations in integral and fractional ex- pressions; the second the solution of equations and problems; the third treats of various subjects which are introduced but rarely into examina- tion papers, and are more briefly discussed. Provision has at the same time been made for the introduction of easy equations and problems at an early stage—for those who prefer such a course. KEY TO ALGEBRA FOR BEGINNERS. Crown 8vo. cloth. s. d. TRIGONOMETRY FOR BEGINNERS. With numerous Examples. New Edition. 18mo. cloth. s. d. Intended to serve as an introduction to the larger treatise on Plane Trigonometry, published by the Author. The same plan has been adopted as in the Algebra for Beginners: the subject is discussed in short chap- ters, and a collection of examples is attached to each chapter. The first fourteen chapters present the geometrical part of Plane Trigonometry; and contain all that is necessary for practical

purposes. The range of matter included is such as seems required by the various examinations in elementary Trigonometry which are now carried on in the country. Answers are appended at the end. MECHANICS FOR BEGINNERS. With numerous Examples. Second Edition. 18mo. cloth. s. d. Intended as a companion to the two preceding books. The work forms an elementary treatise on demonstrative mechanics. It may be true that this part of mixed mathematics has been sometimes made too abstract — —

EDUCATIONAL BOOKS. and speculative; but it can hardly be doubted that a knowledge of the elements at least of the theory of the subject is extremely valuable even for those who are mainly concerned with practical results. The Author has accordingly endeavoured to provide a suitable introduction to the study of applied as well as of theoretical mechanics. The work consists of two parts, namely, Statics and Dynamics. It will be found to contain all that is usually comprised in elementary treatises on Mechanics, together with some additions. ALGEBRA. For the Use of Colleges and Schools. Fifth Edition. Crown 8vo. cloth. s. d. This work contains all the propositions which are usually included in elementary treatises on Algebra, and a large number of Examples for Exercise. The author has sought to render the work easily intelligi- ble to students, without impairing the accuracy of the demonstrations, or contracting the limits of the subject. The Examples, about Sixteen hundred and fifty in number, have been selected with a view to illustrate every part of the subject. Each chapter is complete in itself; and the work will be found peculiarly adapted to the wants of students who are without the aid of a teacher. The Answers to the examples, with hints for the solution of some in which assistance may be needed, are given at the end of the book. In the present edition two New Chapters and Three hundred miscellaneous Examples have been added. The latter are arranged in sets, each set containing ten examples. KEY TO ALGEBRA FOR THE USE OF COLLEGES AND SCHOOLS. Crown 8vo. s. d. AN ELEMENTARY TREATISE ON THE THEORY OF EQUATIONS. Second Edition, revised. Crown 8vo. cloth. s. d. This treatise contains all the propositions which are usually included in elementary treatises on the theory of Equations, together with Ex- amples for exercise. These have been selected from the College and — —

MATHEMATICS. University Examination Papers, and the results have been given when it appeared necessary. In order to exhibit a comprehensive view of the subject, the treatise includes investigations which are not found in all the preceding elementary treatises, and also some investigations which are not to be found in any of them. For the second edition the work has been revised and some additions have been made, the most impor- tant being an account of the researches of Professor Sylvester respecting Newton’s Rule. PLANE TRIGONOMETRY. For Schools and Colleges. Fourth Edition. Crown 8vo. cloth. s. The design of this work has been to render the subject intelligible to beginners, and at the same time to afford the student the opportunity of obtaining all the information which he will require on this branch of Mathematics. Each chapter is followed by a set of Examples: those which are entitled Miscellaneous Examples, together with a few in some of the other sets, may be advantageously reserved by the student for exercise after he has made some progress in the subject. In the Second Edition the hints for the solution of the

Examples have been considerably increased. A TREATISE ON SPHERICAL TRIGONOMETRY. Second Edition, enlarged. Crown 8vo. cloth. s. d. The present work is constructed on the same plan as the treatise on Plane Trigonometry, to which it is intended as a sequel. In the account of Napier's Rules of Circular Parts, an explanation has been given of a method of proof devised by Napier, which seems to have been overlooked by most modern writers on the subject. Considerable labour has been bestowed on the text in order to render it comprehensive and accurate, and the Examples (selected chiefly from College Examination Papers) have all been carefully verified. — — —

EDUCATIONAL BOOKS. PLANE CO-ORDINATE GEOMETRY , as applied to the Straight Line and the Conic Sections. With numerous Examples. Fourth Edition, revised and enlarged. Crown 8vo. cloth. s. d. The Author has here endeavoured to exhibit the subject in a simple manner for the benefit of beginners, and at the same time to include in one volume all that students usually require. In addition, therefore, to the propositions which have always appeared in such treatises, he has introduced the methods of abridged notation, which are of more recent origin; these methods, which are of a less elementary character than the rest of the work, are placed in separate chapters, and may be omitted by the student at first. A TREATISE ON THE DIFFERENTIAL CALCULUS. With numerous Examples. Fourth Edition. Crown 8vo. cloth. s. d. The Author has endeavoured in the present work to exhibit a comprehensive view of the Differential Calculus on the method of limits. In the more elementary portions he has entered into considerable detail in the explanations, with the hope that a reader who is without the assistance of a tutor may be enabled to acquire a competent acquaintance with the subject. The method adopted is that of Differential Coefficients. To the different chapters are appended examples sufficiently numerous to render another book unnecessary; these examples being mostly selected from College Examination Papers. A TREATISE ON THE INTEGRAL CALCULUS AND ITS APPLICATIONS. With numerous Examples. Third Edition, revised and enlarged. Crown 8vo. cloth. s. d. This is designed as a work at once elementary and complete, adapted for the use of beginners, and sufficient for the wants of advanced stu- dents. In the selection of the propositions, and in the mode of estab- lishing them, it has been sought to exhibit the principles clearly, and to — — —

MATHEMATICS. illustrate all their most important results. The process of summation has been repeatedly brought forward, with the view of securing the attention of the student to the notions which form the true foundation of the Calculus itself, as well as of its most valuable applications. Every attempt has been made to explain those difficulties which usually per- plex beginners, especially with reference to the limits of integrations. A new method has been adopted in regard to the transformation of multi- ple integrals. The last chapter deals with the Calculus of Variations. A large collection of exercises, selected from College Examination Papers, has been appended to the several chapters. EXAMPLES OF ANALYTICAL GEOMETRY OF THREE DIMENSIONS. Second Edition, revised. Crown 8vo. cloth. s. A TREATISE ON ANALYTICAL STATICS. With numerous Examples. Third Edition, revised and enlarged. Crown 8vo. cloth. s. d. In this work on statics (treating of the laws of the equilibrium of bodies) will be found all the propositions which usually

appear in trea- tises on Theoretical Statics. To the different chapters examples are appended, which have been principally selected from University Exam- ination Papers. In the Third Edition many additions have been made, in order to illustrate the application of the principles of the subject to the solution of problems. Wilson (J. M.)—ELEMENTARY GEOMETRY. Angles, Parallels, Triangles, Equivalent Figures, the Circle, and Proportion. By J. M. Wilson , M.A., Fellow of St. John’s College, Cambridge, and Mathematical Master in Rugby School. Second Edition. Extra fcap. 8vo. s. d. — —

EDUCATIONAL BOOKS. The distinctive features of this work are intended to be the following. The classification of Theorems according to their subjects; the separation of Theorems and Problems; the use of hypothetical constructions; the adoption of independent proofs where they are possible and simple; the introduction of the terms locus, projection, &c. ; the importance given to the notion of direction as the property of a straight line; the intermixing of exercises, classified according to the methods adopted for their solution; the diminution of the number of Theorems; the compression of proofs, especially in the later parts of the book; the tacit, instead of the explicit, reference to axioms; and the treatment of parallels. ELEMENTARY GEOMETRY. PART II. (separately). The Circle and Proportion. By J. M. Wilson , M.A. Extra fcap. 8vo. s. d. Wilson (W. P.) —A TREATISE ON DYNAMICS. By W. P. Wilson , M.A., Fellow of St. John’s College, Cambridge, and Professor of Mathematics in Queen’s College, Belfast. 8vo. s. d. Wolstenholme.—A BOOK OF MATHEMATICAL PROBLEMS, on Subjects included in the Cambridge Course. By Joseph Wolstenholme , Fellow of Christ’s College, sometime Fellow of St. John’s College, and lately Lecturer in Mathematics at Christ’s College. Crown 8vo. cloth. s. d. Contents: — Geometry (Euclid) — Algebra — Plane Trigonom- etry — Geometrical Conic Sections — Analytical Conic Sections — Theory of Equations — Differential Calculus — Integral Calculus — Solid Geometry — Statics — Elementary Dynamics — Newton — Dy- namics of a Point — Dynamics of a Rigid Body — Hydrostatics — Geometrical Optics — Spherical Trigonometry and Plane Astronomy. — —

SCIENCE. SCIENCE. ELEMENTARY CLASS-BOOKS. The importance of Science as an element of sound education is now generally acknowledged; and accordingly it is obtaining a prominent place in the ordinary course of school instruction. It is the intention of the Publishers to produce a complete series of Scientific Manuals, affording full and accurate elementary information, conveyed in clear and lucid English. The authors are well known as among the foremost men of their several departments; and their names form a ready guarantee for the high character of the books. Subjoined is a list of those Manuals that have already appeared, with a short account of each. Others are in active preparation; and the whole will constitute a standard series specially adapted to the requirements of beginners, whether for private study or for school instruction. ASTRONOMY, by the Astronomer Royal. POPULAR ASTRONOMY . With Illustrations. By G. B. Airy, Astronomer Royal. Sixth and cheaper Edition. 18mo. cloth. s. d. This work consists of six lectures, which are intended “to explain to intelligent persons the principles on which the instruments of an Observatory are constructed (omitting all details, so far as they are merely subsidiary), and the principles on which the observations made with these

instruments are treated for deduction of the distances and weights of the bodies of the Solar System, and of a few stars, omitting all minutiæ of formulæ, and all troublesome details of calculation.” The speciality of this volume is the direct reference of every step to the Observatory, and the full description of the methods and instruments of observation. — □□ —

EDUCATIONAL BOOKS. ASTRONOMY. MR. LOCKYER’S ELEMENTARY LESSONS IN ASTRONOMY. With Coloured Diagram of the Spectra of the Sun, Stars, and Nebulæ, and numerous Illustrations. By J. Norman Lockyer , F.R.S. Seventh Thousand. 18mo. □s. □d. The author has here aimed to give a connected view of the whole subject, and to supply facts, and ideas founded on the facts, to serve as a basis for subsequent study and discussion. The chapters treat of the Stars and Nebulæ; the Sun; the Solar System; Apparent movements of the Heavenly Bodies; the Measurement of Time; Light; the Telescope and Spectroscope; Apparent Places of the Heavenly Bodies; the Real Distances and Dimensions; Universal Gravitation. The most recent astronomical discoveries are incorporated. Mr. Lockyer’s work supplements that of the Astronomer Royal mentioned in the previous article. QUESTIONS ON LOCKYER’S ELEMENTARY LESSONS IN ASTRONOMY. For the use of Schools. By John Forbes-Robertson. 18mo. cloth limp. □s. □d. PHYSIOLOGY. PROFESSOR HUXLEY’S LESSONS IN ELEMENTARY PHYSIOLOGY. With numerous Illustrations. By T. H. Huxley, F.R.S. Professor of Natural History in the Royal School of Mines. Sixteenth Thousand. 18mo. cloth. □s. □d. This book describes and explains, in a series of graduated lessons, the principles of Human Physiology; or the Structure and Functions of the Human Body. The first lesson supplies a general view of the subject. This is followed by sections on the Vascular or Veinous System, and the Circulation; the Blood and the Lymph; Respiration; Sources of Loss and of Gain to the Blood; the Function of Alimentation; Motion and Locomotion; Sensations and Sensory Organs; the Organ of Sight; — □□ —

SCIENCE. the Coalescence of Sensations with one another and with other States of Consciousness; the Nervous System and Innervation; Histology, or the Minute Structure of the Tissues. A Table of Anatomical and Phys- iological Constants is appended. The lessons are fully illustrated by numerous engravings. The manual is primarily intended to serve as a text-book for teachers and learners in boys’ and girls’ schools. QUESTIONS ON HUXLEY’S PHYSIOLOGY FOR SCHOOLS. By T. Alcock , M.D. 18mo. □s. □d. These Questions were drawn up as aids to the instruction of a class of young people in Physiology. BOTANY. PROFESSOR OLIVER’S LESSONS IN ELEMENTARY BOTANY. With nearly Two Hundred Illustrations. Tenth Thousand. 18mo. cloth. □s. □d. This book is designed to teach the Elements of Botany on Professor Henslow’s plan of selected Types and by the use of Schedules. The earlier chapters, embracing the elements of Structural and Physiological Botany, introduce us to the methodical study of the Ordinal Types. The concluding chapters are entitled, “How to dry Plants” and “How to describe Plants.” A valuable Glossary is appended to the volume. In the preparation of this work free use has been made of the manuscript materials of the late Professor Henslow. CHEMISTRY. PROFESSOR ROSCOE’S LESSONS IN ELEMENTARY CHEMISTRY, INORGANIC AND ORGANIC.By Henry E. Roscoe, F.R.S., Professor of Chemistry in Owens College, Manchester. With

numerous Illustrations and Chromo-Litho. of the Solar Spectrum, and of the Alkalies and Alkaline Earths, New Edition . Twenty-sixth Thousand. 18mo. cloth. s. d. — —

EDUCATIONAL BOOKS. It has been the endeavour of the author to arrange the most impor- tant facts and principles of Modern Chemistry in a plain but concise and scientific form, suited to the present requirements of elementary instruction. For the purpose of facilitating the attainment of exactitude in the knowledge of the subject, a series of exercises and questions upon the lessons have been added. The metric system of weights and mea- sures, and the centigrade thermometric scale, are used throughout the work. The new Edition, besides new wood-cuts, contains many additions and improvements, and includes the most important of the latest discoveries. POLITICAL ECONOMY. POLITICAL ECONOMY FOR BEGINNERS. By Millicent G. Fawcett. 18mo. s. d. The following pages have been written mainly with the hope that a short and elementary book might help to make Political Economy a more popular study in boys' and girls' schools. In order to adapt the book especially for school use, questions have been added at the end of each chapter. LOGIC. ELEMENTARY LESSONS IN LOGIC ; Deductive and Inductive, with copious Questions and Examples, and a Vocabulary of Logical Terms. By W. Stanley Jevons , M.A., Professor of Logic in Owens College, Manchester. 18mo. s. d. In preparing these Lessons the author has attempted to show that Logic, even in its traditional form, can be made a highly useful subject of study, and a powerful means of mental exercise. With this view he has avoided the use of superfluous technical terms, and has abstained from entering into questions of a purely speculative or metaphysical character. For the puerile illustrations too often found in works on — —

SCIENCE. Logic, examples drawn from the distinct objects and ideas treated in the natural and experimental sciences have been generally substituted. At the end of almost every Lesson will be found references to the works in which the student will most profitably continue his reading of the subject treated, so that this little volume may serve as a guide to a more extended course of study. PHYSICS. LESSONS IN ELEMENTARY PHYSICS. By Balfour Stewart, F.R.S., Professor of Natural Philosophy in Owens College, Manchester. With numerous Illustrations and Chromoliths of the Spectra of the Sun, Stars, and Nebulæ. 18mo. s. d. A description, in an elementary manner, of the most important of those laws which regulate the phenomena of nature. The active agents, heat, light, electricity, etc., are regarded as varieties of energy, and the work is so arranged that their relation to one another, looked at in this light, and the paramount importance of the laws of energy are clearly brought out. The volume contains all the necessary illustrations, and a plate representing the Spectra of Sun, Stars, and Nebulæ forms a frontispiece. MANUALS FOR STUDENTS. Flower (W. H.) —AN INTRODUCTION TO THE OSTEOLOGY OF THE MAMMALIA. Being the substance of the Course of Lectures delivered at the Royal College of Surgeons of England in . By W. H. Flower , F.R.S., F.R.C.S., Hunterian Professor of Comparative Anatomy and Physiology. With numerous Illustrations. Globe 8vo. s. d. — —

EDUCATIONAL BOOKS. Although the present work contains the substance of a Course of Lectures, the form has been changed, so as the better to adapt it as a handbook for students. Theoretical views have been almost entirely excluded; and while it is impossible in a scientific treatise to avoid the employment of technical terms, it has been the author’s endeavour to use no more than absolutely necessary, and to exercise due care in se- lecting only those that seem most appropriate, or which have received the sanction of general adoption. With a very few exceptions the illus- trations have been drawn expressly for this work from specimens in the Museum of the Royal College of Surgeons. Hooker (Dr.) —THE STUDENT’S FLORA OF THE BRITISH ISLANDS. By J. D. Hooker , C.B., F.R.S., M.D., D.C.L., Director of the Royal Gardens, Kew. Globe 8vo. □□s. □d. The object of this work is to supply students and field-botanists with a fuller account of the Plants of the British Islands than the manu- als hitherto in use aim at giving. The Ordinal, Generic, and Specific characters have been re-written, and are to a great extent original, and drawn from living or dried specimens, or both. Oliver (Professor). —FIRST BOOK OF INDIAN BOTANY. By Daniel Oliver , F.R.S., F.L.S., Keeper of the Herbarium and Library of the Royal Gardens, Kew, and Professor of Botany in University College, London. With numerous Illustrations. Extra fcap. 8vo. □s. □d. This manual is, in substance, the author’s “Lessons in Elementary Botany,” adapted for use in India. In preparing it he has had in view the want, often felt, of some handy r´ esum´ e of Indian Botany, which might be serviceable not only to residents of India, but also to any one about to proceed thither, desirous of getting some preliminary idea of the Botany of that country. — □□ —

SCIENCE. Other volumes of these Manuals will follow. Cooke (Josiah P., Jun.)—FIRST PRINCIPLES OF CHEMICAL PHILOSOPHY. By Josiah P. Cooke , Jun., Ervine Professor of Chemistry and Mineralogy in Harvard College. Crown 8vo. □□s. The object of the author in this book is to present the philosophy of Chemistry in such a form that it can be made with profit the subject of College recitations, and furnish the teacher with the means of testing the student’s faithfulness and ability. With this view the subject has been developed in a logical order, and the principles of the science are taught independently of the experimental evidence on which they rest. Johnson (S. W., M.A.) —HOW CROPS GROW : A Treatise on the Chemical Composition, Structure, and Life of the Plant, for Agricultural Students. By S. W. Johnson , M.A., Professor of Analytical and Agricultural Chemistry in Yale College. With Illustrations and Tables of Analyses. Revised, with Numerous Additions, and adapted for English use by A. H. Church , M.A. and W. T. Dyer , B.A., Professors at the Royal Agricultural College, Cirencester. Crown 8vo. □s. □d. In order that this book may be complete in itself, so far as its spe- cial scope is concerned, not only have the rudiments of Chemistry and structural Botany been introduced, but a series of Experiments has been described, by which the student, who has access to chemical appara- tus and tests, may become conversant with the most salient properties of the elements, and of those of their chief natural compounds, which constitute the food or the materials of plants. It has also been attempted to adapt the work in form and contents to the wants of the class-room by a strictly systematic arrangement of topics, and by division of the matter into convenient paragraphs. — □□ —

EDUCATIONAL BOOKS. Roscoe (H. E.)—SPECTRUM ANALYSIS. Six Lectures, with Appendices, Engravings, Maps, and Chromolithographs. By H. E. Roscoe , F.R.S., Professor of Chemistry in Owens College, Manchester. Royal 8vo. □□s. A Second Edition of these popular Lectures, containing all the most recent discoveries and several additional Illustrations. “The lectures themselves furnish a most admirable elementary trea- tise on the subject, whilst by the insertion in appendices to each lecture of extracts from the most important published memoirs, the author has rendered it equally valuable as a text book for advanced students.”—Westminster Review. Thorpe (T. E.) —A SERIES OF CHEMICAL PROBLEMS, for use in Colleges and Schools. Adapted for the preparation of Students for the Government, Science, and Society of Arts Examinations. With a Preface by Professor Roscoe. 18mo. cloth. □s. In the Preface Dr. Roscoe says—“My experience has led me to feel more and more strongly that by no method can accuracy in a knowledge of chemistry be more surely secured than by attention to the working of well-selected problems, and Dr. Thorpe’s thorough acquaintance with the wants of the student is a sufficient guarantee that this selection has been carefully made. I intend largely to use these questions in my own classes, and I can confidently recommend them to all teachers and students of the science.” Wurtz.—A HISTORY OF CHEMICAL THEORY , from the Age of Lavoisier down to the present time. By Ad. Wurtz . Translated by HENRY WATTS, F.R.S. Crown 8vo. □s. — □□ —

MISCELLANEOUS. MISCELLANEOUS. Abbott.—A SHAKESPEARIAN GRAMMAR. An Attempt to illustrate some of the Differences between Elizabethan and Modern English. By the Rev. E. A. Abbott , M.A., Head Master of the City of London School. For the Use of Schools. New and Enlarged Edition. Extra fcap. 8vo. □s. The object of this work is to furnish students of Shakespeare and Bacon with a short systematic account of some points of difference be- tween Elizabethan syntax and our own. A section on Prosody is added, and Notes and Questions. The success which has attended the First and Second Editions of the “ Shakespearian Grammar,” and the demand for a Third Edi- tion within a year of the publication of the First, has encouraged the Author to endeavour to make the work somewhat more useful, and to render it, as far as possible, a complete book of reference for all difficul- ties of Shakespearian syntax or prosody. For this purpose the whole of Shakespeare has been re-read, and an attempt has been made to include within this Edition the explanation of every idiomatic difficulty (where the text is not confessedly corrupt) that comes within the province of a grammar as distinct from a glossary. The great object being to make a useful book of reference for students, and especially for classes in schools, several Plays have been indexed so fully that with the aid of a glossary and historical notes the references will serve for a complete commentary. ATLAS OF EUROPE. GLOBE EDITION. Uniform in size with Macmillan’s Globe Series, containing 45 Coloured Maps, on a uniform scale and projection: with Plans of London and Paris, and a copious Index. Strongly bound in half-morocco, with flexible back, □s. — □□ —

EDUCATIONAL BOOKS. This Atlas includes all the countries of Europe in a series of 48 Maps, drawn on the same scale, with an Alphabetical Index to the situ- ation of more than ten thousand places; and the relation of the various maps and countries to

each other is defined in a general Key-map. The identity of scale in all the maps facilitates the comparison of extent and distance, and conveys a just impression of the magnitude of different countries. The size suffices to show the provincial divisions, the rail- ways and main roads, the principal rivers and mountain ranges. “This Atlas,” writes the British Quarterly, “will be an invaluable boon for the school, the desk, or the traveller’s portmanteau.” Bates & Lockyer. —A CLASS-BOOK OF GEOGRA- PHY. Adapted to the recent Programme of the Royal Geographical Society. By H. W. Bates , Assistant Secretary to the Royal Geographical Society, and J. N. Lockyer , F.R.A.S. [In the Press. CAMEOS FROM ENGLISH HISTORY. From Rollo to Edward II. By the Author of “The Heir of Redclyffe.” Extra fcap. 8vo. Second Edition, enlarged. s. A Second Series nearly ready. The endeavour has not been to chronicle facts, but to put together a series of pictures of persons and events, so as to arrest the attention, and give some individuality and distinctness to the recollection, by gath- ering together details at the most memorable moments. The “Cameos” are intended as a book for young people just beyond the elementary his- tories of England, and able to enter in some degree into the real spirit of events, and to be struck with characters and scenes presented in some relief. “Instead of dry details,” says the Nonconformist, “we have living pictures, faithful, vivid, and striking.” Delamotte.—A BEGINNER’S DRAWING BOOK. By P. H. Delamotte , F.S.A. Progressively arranged, with upwards of Fifty Plates. Crown 8vo. Stiff covers. s. d. — — —

MISCELLANEOUS. This work is intended to give such instruction to Beginners in Draw- ing, and to place before them copies so easy, that they may not find any obstacle in making the first step. Thenceforward the lessons are gradu- ally progressive. Mechanical improvements too have lent their aid. The whole of the Plates have been engraved by a new process, by means of which a varying depth of tone—up to the present time the distinguishing characteristic of pencil drawing—has been imparted to wood-cuts. D’Oursy and Feillet.—A FRENCH GRAMMAR AT SIGHT, on an entirely new method. By A. D’Oursy and A. Feillet . Especially adapted for Pupils preparing for Examination. Fcap. 8vo. cloth extra. s. d. The method followed in this volume consists in presenting the gram- mar as much as possible by synoptical tables, which, striking the eye at once, and following throughout the same order— “used—not used;” “changes—does not change” —are easily remembered. The parsing ta- bles will enable the pupil to parse easily from the beginning. The exer- cises consist of translations from French into English, and from English into French; and of a number of grammatical questions. EUROPEAN HISTORY. Narrated in a Series of Historical Selections from the Best Authorities. Edited and arranged by E. M. Sewell and C. M. Yonge . First Series, crown 8vo. s. Second Series, – . Crown 8vo. s. When young children have acquired the outlines of History from abridgments and catechisms, and it becomes desirable to give a more enlarged view of the subject, in order to render it really useful and interesting, a difficulty often arises as to the choice of books. Two courses are open, either to take a general and consequently dry history of facts, such as Russel’s Modern Europe, or to choose some work treating of a particular period or subject, such as the works of Macaulay and Froude. The former course usually renders history uninteresting; the — —

EDUCATIONAL BOOKS. latter is unsatisfactory, because it is not sufficiently comprehensive. To remedy this difficulty selections, continuous and chronological, have, in the present volume, been taken from the larger works of Freeman, Milman, Palgrave, and others, which may serve as distinct landmarks of historical reading. “We know of scarcely anything,” says the Guardian, of this volume, “which is so likely to raise to a higher level the average standard of English education.” Freeman (Edward A.)—OLD-ENGLISH HISTORY. By Edward A. Freeman , D.C.L., late Fellow of Trinity College, Oxford. With Five Coloured Maps. New Edition. Extra fcap. 8vo. half-bound. s. “Its object is to show that clear, accurate, and scientific views of history, or indeed of any subject, may be easily given to children from the very first. . . . I have, I hope, shown that it is perfectly easy to teach children, from the very first, to distinguish true history alike from legend and from wilful invention, and also to understand the nature of historical authorities and to weigh one statement against another. . . . I have throughout striven to connect the history of England with the general history of civilized Europe, and I have especially tried to make the book serve as an incentive to a more accurate study of historical geography.”—Preface. Helfenstein (James). —A COMPARATIVE GRAM- MAR OF THE TEUTONIC LANGUAGES. Being at the same time a Historical Grammar of the English Language, and comprising Gothic, Anglo-Saxon, Early English, Modern English, Icelandic (Old Norse), Danish, Swedish, Old High German, Middle High German, Modern German, Old Saxon, Old Frisian, and Dutch. By James Helfenstein , Ph.D. 8vo. s. This work traces the different stages of development through which the various Teutonic languages have passed, and the laws which have — —

MISCELLANEOUS. regulated their growth. The reader is thus enabled to study the relation which these languages bear to one another, and to the English language in particular, to which special attention is devoted throughout. In the chapters on Ancient and Middle Teutonic languages no grammatical form is omitted the knowledge of which is required for the study of an- cient literature, whether Gothic or Anglo-Saxon or Early English. To each chapter is prefixed a sketch showing the relation of the Teutonic to the cognate languages, Greek, Latin, and Sanskrit. Those who have mastered the book will be in a position to proceed with intelligence to the more elaborate works of Grimm, Bopp, Pott, Schleicher, and others. Hole.—A GENEALOGICAL STEMMA OF THE KINGS OF ENGLAND AND FRANCE. By the Rev. C. Hole . On Sheet. s. The different families are printed in distinguishing colours, thus fa- cilitating reference. A BRIEF BIOGRAPHICAL DICTIONARY. Compiled and Arranged by Charles Hole , M.A., Trinity College, Cambridge. Second Edition, 18mo. neatly and strongly bound in cloth. s. d. The inquiry is frequently made concerning an eminent man, when did he live, or for what was he celebrated, or what biographies have we about him? Such information is concisely supplied in this Dictionary. It contains more than 18,000 names. Extreme care has been bestowed on the verification of the dates, and thus numerous errors current in previous works have been corrected. Its size adapts it for the desk, portmanteau, or pocket. “An invaluable addition to our manuals of reference, and from its moderate price cannot fail to become as popular as it is useful.”— Times. — —

EDUCATIONAL BOOKS. Jephson.—SHAKESPEARE'S "TEMPEST." With Glossarial and Explanatory Notes. By the Rev. J. M. Jephson . 18mo. s. d. It is important to find some substitute for classical study, and it is believed that such a substitute may be found in the Plays of Shakespeare. Each sentence of Shakespeare becomes, like a sentence in Thucydides or Cicero, a lesson in the origin and derivation of words, and in the fundamental rules of grammatical construction. On this principle the present edition of the "Tempest" has been prepared. The text is taken from the "Cambridge Shakespeare." M'Cosh (Rev. Principal). —THE LAWS OF DISCURSIVE THOUGHT. Being a Text-Book of Formal Logic. By James M'Cosh , D.D., LL.D. 8vo. s. In this treatise the Notion (with the Term and the Relation of Thought to Language,) will be found to occupy a larger relative place than in any logical work written since the time of the famous "Art of Thinking." Oppen.—FRENCH READER. For the Use of Colleges and Schools. Containing a graduated Selection from modern Authors in Prose and Verse; and copious Notes, chiefly Etymological. By Edward A. Oppen . Fcap. 8vo. cloth. s. d. This is a Selection from the best modern authors of France. Its distinctive feature consists in its etymological notes, connecting French with the classical and modern languages, including the Celtic. This sub- ject has hitherto been little discussed even by the best-educated teachers. A SHILLING BOOK OF GOLDEN DEEDS. A Reading Book for Schools and General Readers. By the Author of "The Heir of Redclyffe." 18mo. cloth. — —

MISCELLANEOUS. A record of some of the good and great deeds of all time, abridged from the larger work of the same author in the Golden Treasury Series. Sonnenschein and Meiklejohn.—THE ENGLISH METHOD OF TEACHING TO READ. By A. Sonnenschein and J. M. D. Meiklejohn , M.A. Fcap. 8vo. Comprising. The Nursery Book , containing all the Two-Letter Words in the Language. d. The First Course , consisting of Short Vowels with Single Consonants. d. The Second Course , with Combinations and Bridges, consisting of Short Vowels with Double Consonants. d. The Third And Fourth Courses , consisting of Long Vowels, and all the Double Vowels in the Language. d. A Series of Books in which an attempt is made to place the process of learning to read English on a scientific basis. This has been done by separating the perfectly regular parts of the language from the irregular, and by giving the regular parts to the learner in the exact order of their difficulty. The child begins with the smallest possible element, and adds to that element one letter—in only one of its functions—at one time. Thus the sequence is natural and complete. Vaughan (C. M.)—A SHILLING BOOK OF WORDS FROM THE POETS. By C. M. Vaughan . 18mo. cloth. It has been felt of late years that the children of our parochial schools, and those classes of our countrymen which they commonly represent, are capable of being interested, and therefore benefited also, by some- thing higher in the scale of poetical composition than those brief and — —

EDUCATIONAL BOOKS. somewhat puerile fragments to which their knowledge was formerly re- stricted. An attempt has here been made to supply the want by forming a selection at once various and unambitious; healthy in tone, just in sentiment, elevating in thought, and beautiful in expression. Thring.—Works by Edward Thring, M.A., Head Master of Uppingham. THE ELEMENTS OF GRAMMAR TAUGHT IN ENGLISH , with Questions. Fourth Edition. 18mo. s. This little work is

chiefly intended for teachers and learners. It took its rise from questionings in National Schools, and the whole of the first part is merely the writing out in order the answers to questions which have been used already with success. A chapter on Learning Language is especially addressed to teachers. THE CHILD’S GRAMMAR. Being the Substance of “The Elements of Grammar taught in English,” adapted for the Use of Junior Classes. A New Edition. 18mo. s. SCHOOL SONGS. A Collection of Songs for Schools. With the Music arranged for four Voices. Edited by the Rev. E. Thring and H. Riccius . Folio. s. d. There is a tendency in schools to stereotype the forms of life. Any genial solvent is valuable. Games do much; but games do not penetrate to domestic life, and are much limited by age. Music supplies the want. The collection includes the “Agnus Dei,” Tennyson’s “Light Brigade,” Macaulay’s “Ivry,” &c. among other pieces. Trench (Archbishop). —HOUSEHOLD BOOK OF ENGLISH POETRY. Selected and Arranged, with Notes, by R. C. Trench , D.D., Archbishop of Dublin. Extra fcap. 8vo. s. d. Second Edition. — — —

MISCELLANEOUS. This volume is called a “Household Book,” by this name implying that it is a book for all—that there is nothing in it to prevent it from being confidently placed in the hands of every member of the household. Specimens of all classes of poetry are given, including selections from living authors. The Editor has aimed to produce a book “which the emigrant, finding room for little not absolutely necessary, might yet find room for it in his trunk, and the traveller in his knapsack, and that on some narrow shelves where there are few books this might be one.” “The Archbishop has conferred in this delightful volume an impor- tant gift on the whole English-speaking population of the world.”—Pall Mall Gazette. Yonge (Charlotte M.). —A PARALLEL HISTORY OF FRANCE AND ENGLAND : consisting of Outlines and Dates. By Charlotte M. Yonge , Author of “The Heir of Redclyffe,” “Cameos of English History,” &c., &c. Oblong 4to. s. d. This tabular history has been drawn up to supply a want felt by many teachers of some means of making their pupils realize what events in the two countries were contemporary. A skeleton narrative has been constructed of the chief transactions in either country, placing a column between for what affected both alike, by which means it is hoped that young people may be assisted in grasping the mutual relation of events. — — —

EDUCATIONAL BOOKS. DIVINITY. Abbott (Rev. E. A.)—BIBLE LESSONS. By the Rev. E. A. Abbott , M.A., Head Master of the City of London School. Second Edition, crown 8vo. s. d. This book is written in the form of dialogues carried on between a teacher and pupil, and its main object is to make the scholar think for himself. The great bulk of the dialogues represents in the spirit, and often in the words, the religious instruction which the author has been in the habit of giving to the Fifth and Sixth Forms of the City of London School. Cheyne (T. K.)—THE BOOK OF ISAIAH CHRONO- LOGICALLY ARRANGED. An Amended Version, with Historical and Critical Introductions and Explanatory Notes. By T. K. Cheyne , M.A., fellow of Balliol College, Oxford. Crown 8vo. s. d. The object of this edition is simply to restore the probable meaning of Isaiah, so far as this can be expressed in modern English. The basis of the version is the revised translation of , but no scruple has been felt in introducing alterations, wherever the true sense of the prophecies appeared to require it. Eastwood.—THE BIBLE WORD-BOOK. A Glossary of Old English Bible

Words. By J. Eastwood , M.A., of St. John’s College, and W. Aldis Wright , M.A., Trinity College, Cambridge. 18mo. s. d. It is the object of this Glossary to explain and illustrate all such words, phrases, and constructions, in the Authorized Version of the Old and New Testaments and the Apocrypha, and in the Book of Common Prayer, as are either obsolete or archaic. Full explanations are supplied, — — —

DIVINITY. and these illustrated by numerous citations from the elder writers. An index of editions quoted is appended. Apart from its immediate subject, this work serves to illustrate a well-marked period in the history of the English language. It is thus of distinct philological value. GOLDEN TREASURY PSALTER. Students’ Edition. Being an Edition of “The Psalms Chronologically Arranged, by Four Friends,” with briefer Notes. 18mo. s. d. In making this abridgment of “The Psalms Chronologically Ar- ranged,” the editors have endeavoured to meet the requirements of readers of a different class from those for whom the larger edition was intended. Some who found the large book useful for private reading, have asked for an edition of a smaller size and at a lower price, for family use, while at the same time some Teachers in Public Schools have suggested that it would be convenient for them to have a simpler book, which they could put into the hands of younger pupils. Hardwick.—A HISTORY OF THE CHRISTIAN CHURCH. Middle Age. From Gregory the Great to the Excommunication of Luther. By Archdeacon Hardwick. Edited by Francis Procter, M.A. With Four Maps constructed for this work by A. Keith Johnston . Second Edition. Crown 8vo. s. d. The ground-plan of this treatise coincides in many points with one adopted at the close of the last century in the colossal work of Schr¨ ockh, and since that time by others of his thoughtful countrymen; but in ar- ranging the materials a very different course has frequently been pur- sued. With regard to the opinions of the author, he is willing to avow distinctly that he has construed history with the specific prepossessions of an Englishman and a member of the English Church. The reader is constantly referred to the authorities, both original and critical, on which the statements are founded. — — —

EDUCATIONAL BOOKS. A HISTORY OF THE CHRISTIAN CHURCH DURING THE REFORMATION. By Archdeacon Hardwick . Revised by Francis Procter, M.A. Second Edition. Crown 8vo. s. d. This volume is intended as a sequel and companion to the “History of the Christian Church during the Middle Age.” The author’s earnest wish has been to give the reader a trustworthy version of those stirring incidents which mark the Reformation period, without relinquishing his former claim to characterise peculiar systems, persons, and events ac- cording to the shades and colours they assume, when contemplated from an English point of view, and by a member of the Church of England. Maclear.—Works by the Rev. G. F. Maclear , B.D., Head Master of King’s College School, and Preacher at the Temple Church. A CLASS-BOOK OF OLD TESTAMENT HISTORY. Fifth Edition, with Four Maps. 18mo. cloth. s. d. This volume forms a Class-book of Old Testament History from the earliest times to those of Ezra and Nehemiah. In its preparation the most recent authorities have been consulted, and wherever it has ap- peared useful, Notes have been subjoined illustrative of the Text, and, for the sake of more advanced students, references added to larger works. The Index has been so arranged as to form a concise dictionary of the persons and places mentioned in the course of the narrative;

while the maps, which have been prepared with considerable care at Stanford’s Geographical Establishment, will, it is hoped, materially add to the value and usefulness of the Book. A CLASS-BOOK OF NEW TESTAMENT HISTORY , including the Connexion of the Old and New Testament. With Four Maps. Third Edition. 18mo. cloth. s. d. — —

DIVINITY. A sequel to the author’s Class-book of Old Testament History, continuing the narrative from the point at which it there ends, and carrying it on to the close of St. Paul’s second imprisonment at Rome. In its preparation, as in that of the former volume, the most recent and trust- worthy authorities have been consulted, notes subjoined, and references to larger works added. It is thus hoped that it may prove at once an useful class-book and a convenient companion to the study of the Greek Testament. A SHILLING BOOK OF OLD TESTAMENT HISTORY , for National and Elementary Schools. With Map. 18mo. cloth. A SHILLING BOOK OF NEW TESTAMENT HISTORY , for National and Elementary Schools. With Map. 18mo. cloth. These works have been carefully abridged from the author’s larger manuals. CLASS-BOOK OF THE CATECHISM OF THE CHURCH OF ENGLAND. Second Edition. 18mo. cloth. s. d. This may be regarded as a sequel to the Class-books of Old and New Testament History. Like them, it is furnished with notes and references to larger works, and it is hoped that it may be found, especially in the higher forms of our Public Schools, to supply a suitable manual of instruction in the chief doctrines of the English Church, and a useful help in the preparation of Candidates for Confirmation. A FIRST CLASS-BOOK OF THE CATECHISM OF THE CHURCH OF ENGLAND , with Scripture Proofs, for Junior Classes and Schools. 18mo. d. THE ORDER OF CONFIRMATION. A Sequel to the Class-book of the Catechism. For the use of Candidates for Confirmation. With Prayers and Collects. 18mo. d. — —

EDUCATIONAL BOOKS. Maurice.—THE LORD’S PRAYER, THE CREED, AND THE COMMANDMENTS. A Manual for Parents and Schoolmasters. By the Rev. F. D. Maurice . To which is added the Order of the Scriptures. 18mo. s. Procter.—A HISTORY OF THE BOOK OF COMMON PRAYER, with a Rationale of its Offices. By Francis Procter, M.A. Ninth Edition, revised and enlarged. Crown 8vo. s. d. In the course of the last twenty years the whole question of Liturgical knowledge has been reopened with great learning and accurate research; and it is mainly with the view of epitomizing extensive publications, and correcting the errors and misconceptions which had obtained currency, that the present volume has been put together. Procter and Maclear.—AN ELEMENTARY INTRO- DUCTION TO THE BOOK OF COMMON PRAYER. Re-arranged and supplemented by an Explanation of the Morning and Evening Prayer and the Litany. By the Rev. F. Procter and the Rev. G. F. Maclear . Fourth Edition. 18mo. s. d. As in the other Class-books of the series, notes have also been sub- joined, and references given to larger works, and it is hoped that the volume will be found adapted for use in the higher forms by our Public Schools, and a suitable manual for those preparing for the Oxford and Cambridge local examinations. This new Edition has been considerably altered, and several important additions have been made. Besides a re-arrangement of the work generally, the Historical Portion has been supplemented by an Explanation of the Morning and Evening Prayer and of the Litany. PSALMS OF DAVID CHRONOLOGICALLY

ARRANGED. By Four Friends . An Amended Version, with Historical — □□ —

DIVINITY. Introduction and Explanatory Notes. Second Edition, with Additions and Corrections. Crown 8vo. □s. □d. To restore the Psalter as far as possible to the order in which the Psalms were written,—to give the division of each Psalm into strophes, of each strophe into the lines which composed it,—to amend the errors of translation, is the object of the present Edition. Professor Ewald’s works, especially that on the Psalms, have been extensively consulted. This book has been used with satisfaction by masters for private work in higher classes in schools. Ramsay.—THE CATECHISER’S MANUAL; or, the Church Catechism illustrated and explained, for the use of Clergymen, Schoolmasters, and Teachers. By the Rev. Arthur Ramsay, M.A. Second Edition. 18mo. □s. □d. A clear explanation of the Catechism, by way of Question and An- swer. Simpson.—AN EPITOME OF THE HISTORY OF THE CHRISTIAN CHURCH. By William Simpson , M.A. Fifth Edition. Fcap. 8vo. □s. □d. A compendious summary of Church History. Swainson.—A HANDBOOK to BUTLER’S ANALOGY. By C. A. Swainson , D.D., Canon of Chichester. Crown 8vo. □s. □d. This manual is designed to serve as a handbook or road-book to the Student in reading the Analogy, to give the Student a sketch or outline map of the country on which he is entering, and to point out to him matters of interest as he passes along. — □□ —

EDUCATIONAL BOOKS. Westcott.—A GENERAL SURVEY OF THE HISTORY OF THE CANON OF THE NEW TESTAMENT DURING THE FIRST FOUR CENTURIES. By Brooke Foss Westcott , B.D., Canon of Peterborough. Third Edition, revised. Crown 8vo. □□s. □d. The Author has endeavoured to connect the history of the New Tes- tament Canon with the growth and consolidation of the Church, and to point out the relation existing between the amount of evidence for the authenticity of its component parts, and the whole mass of Christian literature. Such a method of inquiry will convey both the truest notion of the connexion of the written Word with the living Body of Christ, and the surest conviction of its divine authority. Of this work the Saturday Review writes: “Theological students, and not they only, but the general public, owe a deep debt of gratitude to Mr. Westcott for bringing this subject fairly before them in this candid and comprehensive essay. . . . As a theological work it is at once perfectly fair and impartial, and imbued with a thoroughly religious spirit; and as a manual it exhibits, in a lucid form and in a narrow compass, the results of extensive research and accurate thought. We cordially recommend it.” INTRODUCTION TO THE STUDY OF THE FOUR GOSPELS. By Brooke Foss Westcott , B.D. Third Edition. Crown 8vo. □□s. □d. This book is intended to be an introduction to the Study of the Gospels. The author has made it a point carefully to study the re- searches of the great writers, and consciously to neglect none. There is an elaborate discussion appended “On the Primitive Doctrine of Inspi- ration.” A GENERAL VIEW OF THE HISTORY OF THE ENGLISH BIBLE. By Brooke Foss Westcott , B.D. Crown 8vo. □□s. □d. — □□ —

DIVINITY. “The first trustworthy account we have had of that unique and mar- vellous monument of the piety of our ancestors.”— Daily News. THE BIBLE IN THE CHURCH. A Popular Account of the Collection and Reception of the Holy Scriptures in the Christian Churches. Third Edition. By Brooke Foss Westcott, B.D. 18mo. cloth.

□s. □d. The present book is an attempt to answer a request, which has been made from time to time, to place in a simple form, for the use of general readers, the substance of the author's "History of the Canon of the New Testament." An elaborate and comprehensive Introduction is followed by chapters on the Bible of the Apostolic Age; on the Growth of the New Testament; the Apostolic Fathers; the Age of the Apologists; the First Christian Bible; the Bible Proscribed and Restored; the Age of Jerome and Augustine; the Bible of the Middle Ages in the West and in the East, and in the Sixteenth Century. Two appendices on the History of the Old Testament Canon before the Christian Era, and on the Contents of the most ancient MSS. of the Christian Bible, complete the volume. THE GOSPEL OF THE RESURRECTION. Thoughts on its Relation to Reason and History. By Brooke Foss Westcott, B.D. New Edition. Fcap. 8vo. □s. □d. This Essay is an endeavour to consider some of the elementary truths of Christianity as a miraculous Revelation, from the side of History and Reason. If the arguments which are here adduced are valid, they will go far to prove that the Resurrection, with all that it includes, is the key to the history of man, and the complement of reason. Wilson.—AN ENGLISH, HEBREW, AND CHALDEE LEXICON AND CONCORDANCE , to the more Correct Understanding of the English translation of the Old Testament, by reference to the Original Hebrew. By William Wilson , D.D., Canon of Winchester, late Fellow — □□ —

EDUCATIONAL BOOKS. of Queen's College, Oxford. Second Edition, carefully Revised. 4to. cloth. □□s. The aim of this work is, that it should be useful to clergymen and all persons engaged in the study of the Bible, even when they do not possess a knowledge of Hebrew; while able Hebrew scholars have borne testimony to the help that they themselves have found in it. — □□ —

BOOKS ON EDUCATION. BOOKS ON EDUCATION. Arnold.—A FRENCH ETON; OR, MIDDLE CLASS EDUCATION AND THE STATE. By Matthew Arnold. Fcap. 8vo. cloth. □s. □d. "A very interesting dissertation on the system of secondary in- struction in France, and on the advisability of copying the system in England."—Saturday Review. SCHOOLS AND UNIVERSITIES ON THE CONTINENT. 8vo. □□s. □d. The Author was in□□□□ charged by the Schools Inquiry Commission- ers with the task of investigating the system of education for the middle and upper classes in France, Italy, Germany, and Switzerland. In the discharge of this task he was on the Continent nearly seven months, and during that time he visited the four countries named and made a careful study of the matters to which the Commissioners had directed his attention. The present volume contains the report which he made to them. It is here adapted to the use of the general reader. ESSAYS ON A LIBERAL EDUCATION. Edited by the Rev. F. W. Farrar , M.A., F.R.S., Assistant Master at Harrow, late Fellow of Trinity College, Cambridge, and Hon. Fellow of King's College, London. Second Edition. 8vo. cloth. □□s. □d. Contents:—History of Classical Education, by Charles S. Parker, M.A.; Theory of Classical Education, by Henry Sedgwick, M.A.; Liberal Education in Universities, by John Seeley, M.A.; Teaching by means of Grammar, by E. E. Bowen, M.A.; Greek and Latin Verse-Composition, by the Rev. F. W. Farrar; Natural Science in Schools, by J. M. Wilson, M.A., F.G.S.; The Teaching of English, by J. W. Hales, M.A.; Edu- cation of the Reasoning Faculties, by W. Johnson, M.A.; The present Social Results of Classical Education, by Lord

Houghton. — □□ —

EDUCATIONAL BOOKS. The Authors have sought to hasten the expansion and improvement of liberal education by showing in what light some of the most interesting questions of Educational Reform are viewed by men who have had opportunities for forming a judgment respecting them, and several of whom have been for some time engaged in the work of education at our Universities and Schools. Farrar.—ON SOME DEFECTS IN PUBLIC SCHOOL EDUCATION. A Lecture delivered at the Royal Institution. With Notes and Appendices. Crown 8vo. □s. Jex-Blake.—A VISIT TO SOME AMERICAN SCHOOLS AND COLLEGES. By Sophia Jex-Blake . Crown 8vo. cloth. □s. “In the following pages I have endeavoured to give a simple and accurate account of what I saw during a series of visits to some of the Schools and Colleges in the United States. . . . I wish simply to give other teachers an opportunity of seeing through my eyes what they cannot perhaps see for themselves, and to this end I have recorded just such particulars as I should myself care to know.”— Author’s Preface. “Miss Blake gives a living picture of the Schools and Colleges them- selves in which that education is carried on.”— Pall Mall Gazette. Quain (Richard, F.R.S.) —ON SOME DEFECTS IN GENERAL EDUCATION. By Richard Quain , F.R.S. Crown 8vo. □s. □d. Having been charged by the College of Surgeons with the delivery of the Hunterian Oration for □□□□, the Author has availed himself of the occasion to bring under notice some defects in the general education of the country, which, in his opinion, effect injuriously all classes of the people, and not least the members of his own profession. The earlier pages of the address contain a short notice of the genius and labours of — □□ —

BOOKS ON EDUCATION. John Hunter, but the subject of education will be found to occupy the larger part—from page twelve to the end. Thring.—EDUCATION AND SCHOOL. By the Rev. Edward Thring , M.A., Head Master of Uppingham. Second Edition. Crown 8vo. cloth. □s. □d. Youmans.—MODERN CULTURE : its True Aims and Requirements. A Series of Addresses and Arguments on the Claims of Scientific Education. Edited by Edward L. Youmans, M.D. Crown 8vo. □s. □d. Contents:—Professor Tyndall on the Study of Physics; Dr. Daub- eny on the Study of Chemistry; Professor Henfrey on the Study of Botany; Professor Huxley on the Study of Zoology; Dr. J. Paget on the Study of Physiology; Dr. Whewell on the Educational History of Sci- ence; Dr. Faraday on the Education of the Judgment; Dr. Hodgson on the Study of Economic Science; Mr. Herbert Spencer on Political Ed- ucation; Professor Masson on College Education and Self Education; Dr. Youmans on the Scientific Study of Human Nature. An Appendix contains extracts from distinguished authors, and from the Scientific Evidence given before the Public Schools Commission. — □□ —

LONDON: R. CLAY, SONS, AND TAYLOR, PRINTERS, BREAD STREET HILL.

PROJECT GUTENBERG LICENSE. A End of Project Gutenberg’s Elements of Plane Trigonometry, by Hugh Blackburn